

Practice Test

FORM 0057B

The ACT Assessment®

DIRECTIONS

This booklet contains tests in English, Mathematics, Reading, and Science Reasoning. These tests measure skills and abilities highly related to high school course work and success in college. *CALCULATORS MAY BE USED ON THE MATHEMATICS TEST ONLY.*

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. *DO NOT USE A BALLPOINT PEN.*

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will NOT be penalized for guessing. *IT IS TO YOUR ADVANTAGE TO ANSWER EVERY QUESTION EVEN IF YOU MUST GUESS.*

You may work on each test ONLY when your test supervisor tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may NOT look back to a test on which time has already been called, and you may NOT go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may NOT for any reason fill in ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

DO NOT OPEN THIS BOOKLET UNTIL TOLD TO DO SO.

You may wish to remove this sample answer document from the booklet to use in a practice test session.

ACT ASSESSMENT® 2002-2003 National Answer Sheet

SIDE 1

A NAME, ADDRESS, AND TELEPHONE (Please print.)	
Last Name	First Name
House Number and Street	
City	State
Area Code	Number
ZIP Code	

ALL examinees MUST complete blocks A, B, C, and D.

Registered Examinees: Enter the MATCHING INFORMATION in blocks B, C, and D EXACTLY as it appears on your admission ticket, even if any part of the information is missing or incorrect. Fill in the corresponding ovals. If you do not complete these blocks to match your admission ticket EXACTLY, your scores will be delayed. Leave block E blank.

Standby Examinees: Enter your identifying information in blocks B, C, and D. Fill in the corresponding ovals. Also fill in the Standby Testing oval in block E.

B MATCH NAME (FIRST 8 LETTERS OF LAST NAME)	C SOCIAL SECURITY NUMBER (OR ACT ID NUMBER)	D DATE OF BIRTH
Month	Day	Year
Jan.	Feb.	Mar.
Apr.	May	June
July	Aug.	Sept.
Oct.	Nov.	Dec.

E STANDBY TESTING	
Fill in this oval ONLY if you turned in a standby registration folder at the test center. This will help ACT [®] to match your answer sheet to your registration record.	
☐ Yes, I am testing as a standby examinee.	

ACT[®]

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ONLY REGISTERED Examinees who need to make corrections are to complete any of blocks F, G, H, or I. If you need to make corrections, complete ONLY the block(s) below for which the information on your admission ticket is incomplete or incorrect. Leave the other blocks blank. If the information on your admission ticket is complete and correct, or if you are a standby examinee, DO NOT MARK in blocks F, G, H, or I.

F NAME CORRECTION		
Last Name	First Name	MI
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2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
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8	8	8
9	9	9
0	0	0
A	A	A
B	B	B
C	C	C
D	D	D
E	E	E
F	F	F
G	G	G
H	H	H
I	I	I
J	J	J
K	K	K
L	L	L
M	M	M
N	N	N
O	O	O
P	P	P
Q	Q	Q
R	R	R
S	S	S
T	T	T
U	U	U
V	V	V
W	W	W
X	X	X
Y	Y	Y
Z	Z	Z

G SOCIAL SECURITY NUMBER CORRECTION	H DATE OF BIRTH CORRECTION	I HIGH SCHOOL CODE CORRECTION
Month	Day	Year
Jan.	Feb.	Mar.
Apr.	May	June
July	Aug.	Sept.
Oct.	Nov.	Dec.

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SIDE 2	BOOKLET NUMBER	FORM																																																																	
Test Directions Mark only one oval for each question. Fill in your response completely and neatly. Erase any errors thoroughly. Incorrect Marks: Correct Mark: 	<table border="1" style="margin: auto;"> <tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr><td>7</td><td>8</td><td>9</td><td>0</td><td>1</td><td>2</td></tr> <tr><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr> <tr><td>9</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>0</td></tr> <tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr><td>7</td><td>8</td><td>9</td><td>0</td><td>1</td><td>2</td></tr> <tr><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr> <tr><td>9</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>0</td></tr> </table>	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	<table border="1" style="margin: auto;"> <tr><td> </td><td> </td><td> </td><td> </td></tr> </table>					○ 00578
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TEST 1

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1 00000	11 00000	21 00000	31 00000	41 00000	51 00000	61 00000
2 00000	12 00000	22 00000	32 00000	42 00000	52 00000	62 00000
3 00000	13 00000	23 00000	33 00000	43 00000	53 00000	63 00000
4 00000	14 00000	24 00000	34 00000	44 00000	54 00000	64 00000
5 00000	15 00000	25 00000	35 00000	45 00000	55 00000	65 00000
6 00000	16 00000	26 00000	36 00000	46 00000	56 00000	66 00000
7 00000	17 00000	27 00000	37 00000	47 00000	57 00000	67 00000
8 00000	18 00000	28 00000	38 00000	48 00000	58 00000	68 00000
9 00000	19 00000	29 00000	39 00000	49 00000	59 00000	69 00000

TEST 2

8 000000	16 000000	24 000000	32 000000	40 000000	48 000000	56 000000
1 000000	9 000000	17 000000	25 000000	33 000000	41 000000	49 000000
2 000000	10 000000	18 000000	26 000000	34 000000	42 000000	50 000000
3 000000	11 000000	19 000000	27 000000	35 000000	43 000000	51 000000
4 000000	12 000000	20 000000	28 000000	36 000000	44 000000	52 000000
5 000000	13 000000	21 000000	29 000000	37 000000	45 000000	53 000000
6 000000	14 000000	22 000000	30 000000	38 000000	46 000000	54 000000
7 000000	15 000000	23 000000	31 000000	39 000000	47 000000	55 000000

TEST 3

6 00000	12 00000	18 00000	24 00000	30 00000	36 00000
1 00000	7 00000	13 00000	19 00000	25 00000	31 00000
2 00000	8 00000	14 00000	20 00000	26 00000	32 00000
3 00000	9 00000	15 00000	21 00000	27 00000	33 00000
4 00000	10 00000	16 00000	22 00000	28 00000	34 00000
5 00000	11 00000	17 00000	23 00000	29 00000	35 00000

TEST 4

6 00000	12 00000	18 00000	24 00000	30 00000	36 00000
1 00000	7 00000	13 00000	19 00000	25 00000	31 00000
2 00000	8 00000	14 00000	20 00000	26 00000	32 00000
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4 00000	10 00000	16 00000	22 00000	28 00000	34 00000
5 00000	11 00000	17 00000	23 00000	29 00000	35 00000

DO NOT SIGN THIS UNTIL YOU HAVE COMPLETED THIS ENTIRE SECTION.

I hereby certify that I have truthfully identified myself on this form. I understand that the consequences of falsifying my identity include cancellation of my scores.

Your Signature

Today's Date





ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE." In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Taeko Komiyama's Mountaintop

Taeko Komiyama has one of the most common family names in Japan, so in her case the name is especially fitting. Komiyama translates as

"small mountain view." No other name could of been more appropriate for Ms. Komiyama, who has lived alone on a mountaintop near Tokyo for about fifty years. I visited her there last summer during a trip to see my cousins in Tokyo.

Ms. Komiyama moved to her house in 1948, when she had reached the age of eighteen years old. Although her family wanted her to lead a more traditional life, she was determined to carry on the work of her father, a mountain guide who had died the year before. For her new home she chose to live on the fourth-highest peak in Japan's Hakone range,

1. ☐ A. NO CHANGE
☐ B. which
☐ C. it's
☐ D. but

2. ☐ F. NO CHANGE
☐ G. could be
☐ H. could been
☐ J. would of been

3. ☐ A. NO CHANGE
☐ B. her age was eighteen years old.
☐ C. she had attained eighteen years in age.
☐ D. she was eighteen years old.

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

Mt. Kintoki, which rises 1,213 meters above the sea. 4

4. Given that all of the following sentences are true, which one should the writer add here to help readers visualize the natural landscape that hikers visiting Mt. Kintoki see?

F. The path up Mt. Kintoki passes through plains, then climbs through bamboo groves and pine forests before it reaches the barren summit.
G. The Hakone Mountains are covered with forests, and the path toward Taeko Komiyama's home is a pleasant climb.
H. Climbing up Mt. Kintoki one passes many trees.
J. Mt. Kintoki is located only about two hours from Tokyo, so many tourists climb its slopes on weekends, and a lesser number make the ascent on weekdays.

5. A. NO CHANGE
B. Hikers typically find
C. Hikers typically, find
D. Hikers, typically find

6. F. NO CHANGE
G. Komiyama; however
H. Komiyama however,
J. Komiyama, however;

7. A. NO CHANGE
B. summit
C. summit's top
D. summit by getting there

8. F. NO CHANGE
G. welcome:
H. welcome; since
J. welcome, although,

9. At this point, the writer is considering adding the following sentence:

The criminal had escaped from a jail in the tiny town below Komiyama's mountain; he had lived a hard and lonely life on the streets before he was imprisoned for robbery.

Would this be a relevant addition at this point in the essay?

- A. Yes, because it helps readers understand just how dangerous this particular "visitor" was.
B. Yes, because knowing that there is a prison below Komiyama's mountain helps the reader visualize the mountain.
C. No, because the details of the criminal's life and of his particular crimes are not of importance to the overall essay.
D. No, because the sentence gives details that should only be used in fictional writing.

Hikers, typically, find, that the climb takes at least three
5

hours. Taeko Komiyama, however, can reach the
6

summit attaining it in only forty minutes.
7

Life on a mountaintop may seem like a lonely undertaking, but Ms. Komiyama hasn't ever lacked for visitors. Her guest book is filled with names from all over the world. Even a prime minister of Japan has made the climb to her simple home. Not all visitors, however, have been so welcome, years ago, an escaped convict made his way to her mountain

retreat and threatened her. 9 Ms. Komiyama,

1 1 1 1 1 1 1 1 1 1 1

whose skilled at judo, threw the intruder across the room.
10

He fled but they recaptured and imprisoned.
11

These days life on the mountain is less eventful. Mt. Kintoki is a popular hiking spot, and "Kintoki's daughter," as Ms. Komiyama has come to be known, greets every person who crests the slope with a friendly smile and a cup of steaming tea. 12 White clouds mist up from the sea,

you can see sunlight shimmering on the snowy slopes of
13

Mt. Fuji. From Mt. Kintoki, it isn't hard to understand
14

what has kept Taeko Komiyama there for all these years.
15

10. F. NO CHANGE
G. whose skilled in
H. who's skilled of
J. who's skilled in

11. A. NO CHANGE
B. recaptured and imprisoned him.
C. was recaptured and imprisoned.
D. they sent him back to prison after being recaptured.

12. Given that all the following sentences are true, which one provides material most relevant to what follows?
F. I imagine she thinks of her father when tourists in thick socks and heavy boots appear.
G. The tea is black and is brewed in a special pot reserved for visitors.
H. This is a tradition she began many years ago, shortly after the criminal was imprisoned.
J. It is wonderful to sip this tea and gaze out across the volcanic landscape.

13. A. NO CHANGE
B. also shimmering sunlight
C. and sunlight shimmers
D. and a shimmering sunlight

14. F. NO CHANGE
G. its not
H. its' not
J. there isn't

15. Assuming that all the choices are true, which one most effectively concludes the sentence and the essay?
A. NO CHANGE
B. why Taeko Komiyama misses her father so much.
C. why Taeko Komiyama enjoys being host to so many visitors.
D. why Taeko Komiyama has become so well known.

PASSAGE II

Public Readings

One popular practice among students in many colleges has been for participation in a marathon oral reading of an important literary work. A professor would assign the work to be read, arrange for a location, and schedule the readers. James Joyce's monumental novel *Ulysses* was always a natural choice, not only because of its significance but also because it details twenty-four

16. F. NO CHANGE
G. by participating
H. as participants
J. to participate

hours in the life of the main character. ¹⁷

A recent trend among college students involve their
¹⁸
own organizing of public readings of works ranging
from John Milton's seventeenth-century epic
poem *Paradise Lost* to Toni Morrison's recent
prize-winning novel *Beloved*. ¹⁹ Interested students

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can sign up for say, half-hour shifts of reading aloud, and
²⁰
many students come—bringing pillows and pizzas—and
manage to stay around to hear the entire book.

Though possibly being a reminder of campus
²¹

fads of years past, as such packing
²²
Volkswagen Bugs, it comes as good news.

Reading would have long been regarded as a private act,
²³

a deed of an individual and personal nature best
²⁴
undertaken in silence. People often think of reading as a
way to escape from the concerns of the world. But reading

17. The writer is considering adding the following true statement to the end of the preceding sentence (replacing the period after the word *character*):

—and that's about how long it takes to read the book.

Would this be a relevant addition to the paragraph?

- A. Yes, because it explains why the students choose not to read the book on their own.
B. Yes, because it offers an interesting detail about the book that logically fits here.
C. No, because it fails to provide any information about the story line of *Ulysses*.
D. No, because a statement this general and vague is not appropriate to this essay.
18. F. NO CHANGE
G. involves their
H. involve they're
J. involves they're
19. The writer is thinking of deleting the phrases "seventeenth-century epic poem" and "recent prize-winning novel" from the preceding sentence. If this were done, the sentence would primarily lose:
- A. facts that support the point that these books are very popular on college campuses.
B. information that describes the range of books being used for these public readings.
C. the confusion caused by the attempted comparison of books from different times and genres.
D. redundant descriptions because they have been mentioned elsewhere in the essay.
20. F. NO CHANGE
G. up for, say
H. up, for say,
J. up for, say,
21. A. NO CHANGE
B. it maybe reminds us
C. this trend might remind us
D. we could be reminded
22. F. NO CHANGE
G. passing, like
H. passing, as
J. past, like
23. A. NO CHANGE
B. If reading had been long
C. Reading has long been
D. Suppose reading were long
24. F. NO CHANGE
G. something best attempted to be
H. a project of activity best
J. one best

aloud in public is an activity worth supporting, for a number of reasons. Otherwise, it's an effective way to bring people together. Nearly every public ritual in our lives include someone reading aloud from a shared text. Such is the ceremonial power we give to the spoken text that we just don't think of each other as properly married or buried; until the right words from a book are read. Likewise, a judge reads the oath of office during the inauguration of an elected official.

Public readings also have value from an aesthetic standpoint. The words of a novel or poem are literally "bound" within there pages. Reading silently and alone, we are apt to neglect the dynamics of cadence and tone that give rise to some of the more enjoyable moments in literature. Reading books aloud and hearing them read can help release the rhythms and eloquence present in the work.

25. A. NO CHANGE
B. At the same time,
C. First,
D. Thus,
26. F. NO CHANGE
G. include somebody
H. includes someone
J. includes them
27. A. NO CHANGE
B. buried until
C. buried. Until when
D. buried if
28. F. NO CHANGE
G. an aesthetic standpoint of view.
H. the view of an aesthetic standpoint.
J. the outlook of an aesthetic viewpoint.
29. A. NO CHANGE
B. its
C. one's
D. these
30. F. NO CHANGE
G. rhythms, and eloquence
H. rhythms and eloquence.
J. rhythms, and, eloquence

PASSAGE III

Bursting Bubbles

Bubble gum is more elastic than its older sibling, chewing gum. Both are made from an insoluble plastic material, but bubble gum has a firmer base, which produces a substance strong enough to hold air while it stretches to create large bubbles.

The first bubble gum, Blibber-Blubber, was developed by the Fleer Corporation in 1906, but it proved to be too sticky and brittle to be successfully marketed. In August of 1928, Walter Diemer, a young accountant employed by the

31. A. NO CHANGE
B. gum, is more elastic than
C. gum, is more elastic, than
D. gum is more elastic than,
32. F. NO CHANGE
G. to
H. much to
J. OMIT the underlined portion.



Fleer Corporation, stumbled upon the first promising formula for bubble gum. ³³ Diemer had no knowledge of

33. At this point, the writer is considering adding the following sentence:

In the latter part of the nineteenth century, chewing gum manufacturers had adopted chicle, the latex of the sapodilla evergreen tree used in the Yucatán Peninsula for centuries as a breath freshener, as a base for chewing gum.

Would this be a relevant addition to make here?

- A. Yes, because the essay explores the origins of bubble gum manufacturing.
- B. Yes, because Diemer had to replace chicle with a product that would stretch more to allow bubble blowing.
- C. No, because the essay focuses on the origins of bubble gum, not chewing gum.
- D. No, because the essay focuses on the popularity of bubble gum, not on its origins or manufacturing.

34. The writer wants to emphasize the characterization that Diemer "stumbled upon" his discovery. Given this purpose, which choice would work best?

- F. NO CHANGE
- G. by trial and error
- H. against the clock
- J. OMIT the underlined portion.

35. A. NO CHANGE
B. formula that he discovered
C. formula he discovered
D. formula

36. Given that all of the following sentences are true, which one should be placed here to offer a logical explanation for why Diemer made a second large batch of gum?

- F. Fellow employees had been afraid that such a large, undiluted batch would break the blades of the mixing machine.
- G. Unfortunately, the gray mass did not look appetizing.
- H. Diemer felt very self-conscious, since all of his co-workers were standing around looking skeptical.
- J. Diemer's formula has been called "The King of Bubble Gum."

37. A. NO CHANGE
B. Most
C. For example,
D. Otherwise most

38. F. NO CHANGE
G. named its
H. called they're
J. called it's

chemistry but had been working ³⁴ enthusiastically for more than a year. Four months after his discovery, the

formula, which he had discovered was nearly perfected and
³⁵

the first large batch of bubble gum was made. ³⁶ The next day, when he made the second large batch, Diemer added a touch of red food coloring because that was all he had.

In other words, most bubble gum has been pink ever since.
³⁷

The Fleer Corporation named it's new product Dubble
³⁸ Bubble. It was test-marketed at a small candy store in

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

Philadelphia. Dubble Bubble sold quick as a wink and was
an instant success. Although Walter Diemer, now a retired

senior vice president of the Flee Corporation never

patented his invention, while he had hoped to keep his
recipe a secret from other gum manufacturers. However,
twenty other companies soon discovered the recipe and
began competing for a share of the market.

Today, the bubble gum business is a major
international industry. Japan, for example, manufactures
more than 150 flavors of bubble gum, including pickled
plum. Flee Corporation, now only one of countless bubble
gum manufacturers, makes four million pieces of bubble

gum a day selling them in fifty countries. In the United
States alone, sales are estimated to be \$150 million a year,
more than the nation spends on elementary school
textbooks. Even astronauts orbiting the earth chew sugar-
free bubble gum while in orbit. Walter Diemer transformed
an unappetizing gray mass into innumerable sweet pink
bubbles that are enjoyed all over the world, and beyond.

39. A. NO CHANGE
B. was an overnight success and
C. sold out in the blink of an eye
D. OMIT the underlined portion.
40. F. NO CHANGE
G. president, of the Flee Corporation,
H. president of the Flee Corporation,
J. president, of the Flee Corporation
41. A. NO CHANGE
B. and
C. so
D. OMIT the underlined portion.

42. F. NO CHANGE
G. Corporation, now,
H. Corporation now,
J. Corporation: now,

43. A. NO CHANGE
B. as they are sold
C. and sells them
D. that is sold

44. F. NO CHANGE
G. during their time in space.
H. which does not have sugar in it.
J. OMIT the underlined portion and end the sentence with a period.

Question 45 asks about the preceding passage as a whole.

45. Suppose the writer had chosen to write an essay that explored the marketing strategies currently used by a major company to promote a product. Would this essay successfully fulfill the writer's goal?
- A. No, because there is not a detailed discussion of current marketing strategies by any single gum company.
B. No, because the essay doesn't mention the role of marketing or competition in the history of bubble gum.
C. Yes, because the essay shows that while Flee's original bubble gum was not successful, Diemer's version, because it was properly marketed, was.
D. Yes, because the essay explores the competition among gum companies for recipes in order to win customers.

PASSAGE IV

The following paragraphs may or may not be in the most logical order. Each paragraph is numbered in brackets, and question 60 will ask you to choose where Paragraph 3 should most logically be placed.

Hot as You Like It

[1]

A finger jabbed my arm as I waited for my order at a taco stand in Santa Fe. "You don't want to order it hot," a man I didn't know due to the fact we hadn't met

insisted, he flags down the stand owner. "Make his order

mild," he told her. "He's got Maine plates on his car." [48]

[2]

While most vegetables taste similar from one species of the plant to another, the chili pepper, backbone of the incendiary cuisine of the Southwest, varies immensely. Each species being a different size, shape, color, and heat.

The pungency, luckily for chili lovers, is rated in Scoville Heat Units (SHU).

46. F. NO CHANGE
G. I had never met
H. of whom I had not been introduced
J. hitherto before I had not made the acquaintance of
47. A. NO CHANGE
B. flagging
C. flags
D. he is flagging
48. If the writer deletes the phrase "in Santa Fe" from the first sentence and also deletes the final sentence of this paragraph, the essay would lose details that are:
F. vivid but not essential to understanding the anecdote described in the paragraph.
G. confusing because of the great geographical distance represented by the two places.
H. helpful to understanding the stranger's actions, which would otherwise be unclear.
J. humorous and thought-provoking, since they indicate the narrator's probable preference for hot food.
49. A. NO CHANGE
B. Once
C. Certainly
D. Because
50. F. NO CHANGE
G. having
H. has
J. OMIT the underlined portion.
51. A. NO CHANGE
B. pungency, luckily for chili lovers
C. pungency luckily for chili lovers,
D. pungency, luckily for chili lovers:



The Scoville scale can be found cited in reference books.

52

For example, the chili used extensively in the

Southwest is the Anaheim pepper, it's fairly mild,

53

with an SHU rating of 1,000 to 3,500. The Anaheim pepper often graces doorways tied into ropes called *ristras*.

54

Then there's the Scotch bonnet, which ought to be called the scorch bonnet, since one walnut-sized pepper can heat a vat of salsa, with an SHU rating of 300,000.

[3]

I was too surprised at the stranger's audacity to object, but it was soon, little time had passed, that as I forked up my burrito, tears of gratitude streamed down my face. That chili sauce was the hottest that had ever passed my lips mingled with tears of pain. What's mild in the Southwest was to my tongue blistering hot. Obviously, the hot sauce I usually gulped without hesitation at home was not the same as what I had just swallowed.

[4]

A little information can be a dangerous thing. Since I had learned the Anaheim pepper is rated mild, I took a big bite. My mouth was burning so much I could have swallowed hot coals to cool it down, proving that a chili is hard to classify objectively. Factors such as soil type or the

52. The writer wishes to dramatize Scoville Heat Units and clarify how they are useful to a person eating chili peppers. Which choice best accomplishes both purposes?

- F. NO CHANGE
- G. Makers of bottled chili sauce use the Scoville scale to help them make a delicious product.
- H. The Scoville scale is like the Richter scale that's used to measure earthquakes: you'll know the damage a chili can do before biting it.
- J. The Scoville scale was invented at the turn of the century: the rating is determined by isolating volatile oils in a lab.

53. A. NO CHANGE

- B. pepper; its
- C. pepper, its
- D. pepper: it's

54. F. NO CHANGE

- G. Tied into ropes called *ristras*, the doorways are often graced with the Anaheim pepper.
- H. The Anaheim pepper tied into ropes, often graces doorways, called *ristras*.
- J. Tied into ropes called *ristras*, the Anaheim pepper often graces doorways.

55. A. NO CHANGE

- B. theirs
- C. they're
- D. there are

56. F. NO CHANGE

- G. not long after, it was soon when
- H. before long, right away
- J. before long,

57. A. NO CHANGE

- B. Then tears of pain, the hottest that had ever passed my lips, were with the hottest chili sauce mingled.
- C. Mingled with those were tears of pain: that sauce was the hottest that had ever passed my lips.
- D. That sauce was the hottest that had ever passed my lips: mingled with those were tears of pain.

1 amount of rainfall or sunshine can affect the heat. Even the parts of a chili ⁵⁸ vary the seeds and veins of a pepper are hotter than the flesh.

[5]

These days I enjoy hot peppers, but one was still ⁵⁹ surprised occasionally by their bite. Unpredictability is part of the pleasure of adding chilies to my food.

58. F. NO CHANGE
G. vary;
H. vary.
J. vary but

59. A. NO CHANGE
B. one still is
C. you still are
D. I am still

Question 60 asks about the preceding passage as a whole.

60. For the sake of the logic and coherence of the essay, Paragraph 3 should be placed:
F. where it is now.
G. before Paragraph 1.
H. after Paragraph 1.
J. after Paragraph 4.

PASSAGE V

A Voice of Her Own

Sandra Cisneros, perhaps the best known Latina author in the United States, ⁶¹ writes poems and stories whose titles alone—"Barbie-Q," "My Lucy Friend Who Smells Like Corn," "Woman Hollering Creek"—engage potential readers' curiosity. ⁶²

Ironically, this renowned writer, whose books are printed on recycled paper, ⁶³ did not do well in school. When she lectures at schools and public libraries, Cisneros presents the evidence. ⁶⁴ An elementary school report card containing Cs, Ds, and a solitary B (for conduct). Cisneros has a theory to explain her low grades: teachers had low expectations for Latina and Latino students from Chicago's South Side.

61. A. NO CHANGE
B. author and writer
C. author and novelist
D. wordsmith and author
62. F. NO CHANGE
G. potential, reader's
H. potential, readers
J. potential readers
63. A. NO CHANGE
B. writer, who is recognized by her orange and black eyeglasses,
C. writer, who likes to write at night,
D. writer
64. F. NO CHANGE
G. evidence: an
H. evidence; an
J. evidence an



Despite the obstacles that she faced in school,
Cisneros completed not only high school but also college.
Her persistence paid off in her twenties, when Cisneros
was admitted prestigious to the Writers' Workshop at the
University of Iowa.

Cisneros soon observed that most of her
classmates at the university seemed to have a common
set of memories, based on middle-class childhoods,
from which to draw in their writing. Cisneros

felt decided out of place.

69 She decided to speak from her own experience.

Her voice, which by being one of a Latina living outside
the mainstream, found a large and attentive audience in

1984 with the publication of her first short story collection,
The House on Mango Street. Today, this book is read by
middle school, high school, and college students across the
United States.

Cisneros uses her influence as a successful writer to
help other Latina and Latino writers get their works
published. But having made the argument that, in order for
large numbers of young Latinos to achieve literary success,

65. The best placement for the underlined portion would be:
- A. where it is now.
 - B. before the word *admitted*.
 - C. before the word *Writers'*.
 - D. before the word *Workshop*.
66. F. NO CHANGE
G. furthermore
H. nevertheless
J. therefore
67. A. NO CHANGE
B. Cisneros herself,
C. Cisneros, herself
D. Cisneros,
68. F. NO CHANGE
G. deciding
H. decidedly
J. decidedly and
69. Which of the following true statements, if added here, would best serve as a transition between the challenges Cisneros faced as an aspiring writer and her success in meeting those challenges?
- A. She did not know what to do.
 - B. Then she had a breakthrough.
 - C. At that point she almost went home to Chicago.
 - D. She wondered whether she was in the right field.
70. F. NO CHANGE
G. voice—that of a Latina living outside the mainstream—
H. voice, being one of a Latina living outside the mainstream, it
J. voice—in which it was a Latina living outside the mainstream—
71. A. NO CHANGE
B. 1984. With
C. 1984; with
D. 1984, with,
72. F. NO CHANGE
G. she argues that,
H. arguing that,
J. she argues that, when

1 1 1 1 1 1 1 1 1 1 1

the educational system itself must change. Cisneros ⁷³hints
that she succeeded in spite of the educational system. "I'm

the exception," she insists, "not the rule." 74

73. Which choice best shows that Cisneros is emphatic about expressing the belief stated in this sentence?

- A. NO CHANGE
- B. says
- C. supposes
- D. asserts

74. The writer is considering deleting the preceding sentence. If the writer decided to delete this sentence, the paragraph would primarily lose a statement that:

- F. enhances the subject and the setting.
- G. provides support for a point previously made.
- H. humorously digresses from the main topic of the paragraph.
- J. contradicts Cisneros's claim made earlier in the essay.

Question 75 asks about the preceding passage as a whole.

75. The writer is considering adding the following sentence to the end of the first paragraph:

To the pleasure of her readers, Cisneros's work, which uses both English and Spanish, is as interesting as the titles suggest.

Should the writer make this addition?

- A. Yes, because it expands on the description of Cisneros's work beyond the mention of the titles.
- B. Yes, because this is the only place in the essay where Cisneros's readers' pleasure is mentioned.
- C. No, because the essay does not mention elsewhere that Cisneros uses English and Spanish in her work.
- D. No, because the writer cannot prove that most readers of Cisneros's work enjoy the titles.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

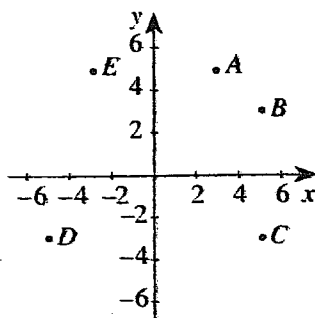
You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. Which point in the standard (x,y) coordinate plane below has the coordinates $(-3,5)$?



- A. A
B. B
C. C
D. D
E. E

2. A bag contains 4 red jelly beans, 5 green jelly beans, and 3 white jelly beans. If a jelly bean is selected at random from the bag, what is the probability that the jelly bean selected is green?

- F. $\frac{1}{12}$
G. $\frac{1}{5}$
H. $\frac{5}{23}$
J. $\frac{5}{12}$
K. $\frac{5}{7}$

3. The balance in Joan's savings account tripled during the year. Joan then withdrew \$500, and the resulting balance was \$100. What was the balance in the account before it tripled?

- A. \$200
B. \$300
C. \$400
D. \$500
E. \$600

4. For what value of x is the equation $2(x - 6) + x = 18$ true?

- F. 15
G. 10
H. 8
J. 4
K. 2

5. An earring manufacturing company has fixed costs of \$10,000 per month and production costs of \$0.60 for each pair of earrings it makes. If the company produces x pairs of earrings in a month, which of the following expressions represents the total of the company's monthly costs?

- A. $10,000x$
B. $10,000 + x$
C. $10,000x + \$0.60$
D. $10,000 + \$0.60x$
E. $(10,000 + \$0.60)x$

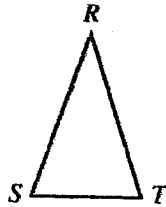
6. Anton went to Mexico during summer vacation with his Spanish class. He recorded the number of pesos he spent each day in a table, as shown below. What was the mean number of pesos he spent per day?

July	1	2	3	4	5
Pesos spent	250	100	150	100	400

- F. 100
G. 150
H. 200
J. 220
K. 300



7. In $\triangle RST$, shown below, $\overline{RS} \cong \overline{RT}$, and the measure of $\angle R$ is 40° . What is the measure of $\angle S$?



- A. 20°
B. 40°
C. 50°
D. 70°

E. Cannot be determined from the given information

8. In the formula $d = rt$, d represents distance in miles, r represents average speed in miles per hour, and t represents time in hours. What is an automobile's average speed, in miles per hour, if it travels 60 miles in $1\frac{1}{2}$ hours?

- F. 30
G. 40
H. 60
J. 90
K. 120

9. $|3 - 2| - |1 - 4| = ?$

- A. -4
B. -2
C. 2
D. 4
E. 10

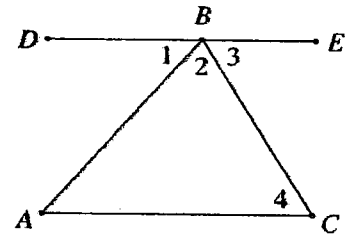
10. If c , d , and f are nonzero real numbers and $cd = f$, which of the following equations for c must always be true?

- F. $c = df$
G. $c = \frac{d}{f}$
H. $c = \frac{f}{d}$
J. $c = f - d$
K. $c = \sqrt{df}$

11. A cookie recipe calls for $\frac{2}{3}$ cup sugar to make 24 two-inch cookies. According to this recipe, how many cups of sugar should be used to make 60 two-inch cookies?

- A. $\frac{5}{9}$
B. 1
C. $1\frac{1}{5}$
D. $1\frac{2}{3}$
E. $3\frac{1}{6}$

12. In the figure below, B is on $\overline{DE}$ and $\overline{DE} \parallel \overline{AC}$. Which of the following angle congruences must hold?



- F. $\angle 1 \cong \angle 2$
G. $\angle 1 \cong \angle 4$
H. $\angle 2 \cong \angle 3$
J. $\angle 2 \cong \angle 4$
K. $\angle 3 \cong \angle 4$

13. For what value of a is $x = 3$ a solution to the equation $x + 3 = ax + 9$?

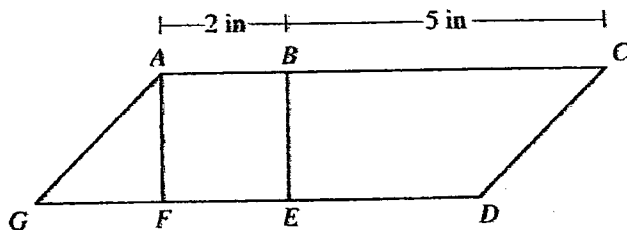
- A. 1.5
B. 1
C. -1
D. -1.5
E. -3

14. Quadrilateral $ABCD$ has vertices $(-2, -1)$, $(4, -3)$, $(5, 2)$, and $(-1, 3)$ in the standard (x, y) coordinate plane. Suppose $ABCD$ is translated 2 units to the left and 1 unit down, forming quadrilateral $A'B'C'D'$. Which of the following shows the coordinates of the vertices of $A'B'C'D'$?

- F. $(-4, -2)$, $(2, -4)$, $(3, 1)$, $(-3, 2)$
G. $(-3, -3)$, $(3, -5)$, $(4, 0)$, $(-2, 1)$
H. $(-2, -2)$, $(4, -4)$, $(5, 1)$, $(-1, 2)$
J. $(0, 0)$, $(6, -2)$, $(7, 3)$, $(1, 4)$
K. $(4, 2)$, $(-8, 6)$, $(-10, -4)$, $(2, -6)$

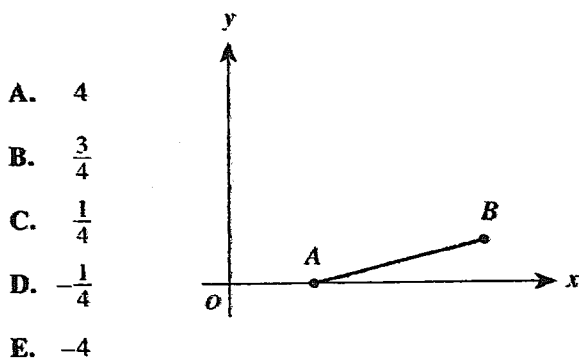


15. In the figure below, F and E are points on $\overline{GD}$. What is the ratio of the area of square $ABEF$ to the area of parallelogram $ACDG$?



- A. 1:14
B. 1:7
C. 7:1
D. 7:2
E. 2:7
16. When $a = b$ and $c = d$, which of the following equations must be true?
F. $a + b = c + d$
G. $a + d = b + c$
H. $a + c = a + b$
J. $a - c = d - b$
K. $ad = cd$

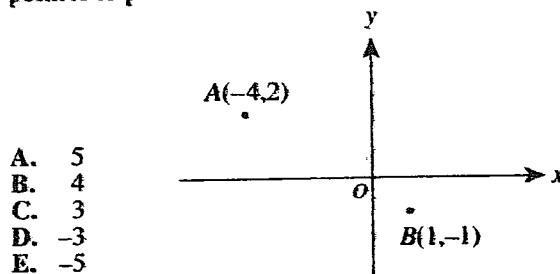
17. The scales on both axes of the standard (x,y) coordinate plane below are the same. Of the following, which is the best estimate for the slope of $\overline{AB}$?



18. A road map is drawn to scale so that 1.5 inches represents 90 miles. How many miles does 1.6 inches represent?

F. 91
G. 96
H. 99
J. 100
K. 106

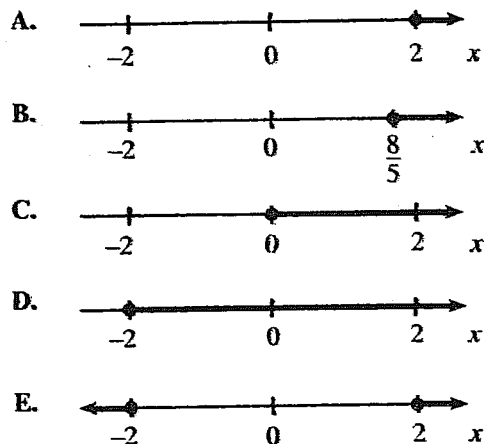
19. In the standard (x,y) coordinate plane shown below, what is the distance in the x direction, in units, from point A to point B ?



20. Which of the following is NOT a solution of $(x-3)(x-1)(x+3)(x+7) = 0$?

F. -7
G. -3
H. 1
J. 3
K. 7

21. Which of the following shows the solution set for the inequality $5x - 1 \geq 9$?



22. If $a = 10$, then which of the following represents 8,003?

F. $8a + 3$
G. $80a + 3$
H. $8a^2 + 3$
J. $8a^3 + 3$
K. $8a^4 + 3$



23. What is the value of b in the solution to the system of equations below?

$$\begin{aligned} 3a - 2b &= 21 \\ a + 3b &= -4 \end{aligned}$$

- A. 17
B. 9
C. 5
D. -3
E. -5

24. Which of the following is an equivalent form of $x + x(x + x)$?

- F. $4x$
G. $x^2 + 2x$
H. $2x^2 + x$
J. $4x^2$
K. x^4

25. To check the slope of a ramp, a building inspector places an overlay of the standard (x,y) coordinate plane on the construction blueprint so that the x -axis aligns with the horizontal on the blueprint. The line segment representing the side view of the ramp goes through the points $(1,-3)$ and $(14,2)$. What is the slope of the planned ramp?

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- A. $-\frac{1}{15}$
B. $-\frac{1}{13}$
C. $-\frac{1}{6}$
D. $\frac{5}{13}$
E. $\frac{13}{5}$

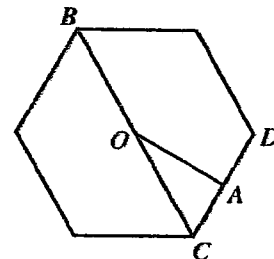
26. Due to inflation, a car that formerly sold for \$15,000 now sells for 10% more. Which of the following calculations gives the current cost, in dollars, of the car?

- F. $15,000 + 10$
G. $15,000 + 15,000(0.01)$
H. $15,000 + 15,000(0.10)$
J. $15,000 + 15,000(10)$
K. $15,000(0.10)$

27. In 3-dimensional space, the set of all points 12 units from the origin is:

- A. a circle.
B. a sphere.
C. a line.
D. a cylinder.
E. 2 parallel planes.

28. In the regular hexagon below, vertices B , C , and D are labeled; $\overline{OA}$ is perpendicular to $\overline{CD}$; A is the midpoint of $\overline{CD}$; and O is the midpoint of $\overline{BC}$. What is the degree measure of $\angle AOC$?



- F. 15°
G. 20°
H. 30°
J. 45°
K. 60°

29. The number 0.005 is 100 times as large as which of the following numbers?

- A. 0.5
B. 0.05
C. 0.0005
D. 0.00005
E. 0.000005

30. The volume, V , of a sphere is determined by the formula $V = \frac{4\pi r^3}{3}$, where r is the radius of the sphere.

What is the volume, in cubic centimeters, of a sphere with a diameter 6 centimeters long?

- F. 36π
G. 72π
H. 108π
J. 144π
K. 288π

31. Which of the following is equal to $\frac{\frac{1}{2} - \frac{1}{3}}{\frac{1}{2} + \frac{1}{3}}$?

- A. $-\frac{5}{1}$
B. $-\frac{1}{5}$
C. $\frac{1}{2}$
D. $\frac{1}{5}$
E. $\frac{1}{6}$

32. One neon sign flashes every 6 seconds. Another neon sign flashes every 8 seconds. If they flash together and you begin counting seconds, how many seconds after they flash together will they next flash together?

- F. 48
G. 24
H. 14
J. 7
K. 2



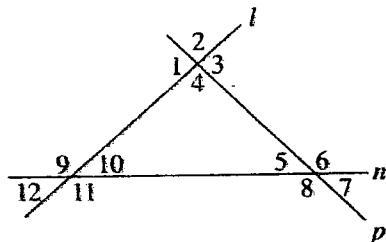
33. If $\sqrt{2x} - 5 = 1$, then $x = ?$

- A. -8
- B. 8
- C. 9
- D. 12
- E. 18

34. A truck sprang a leak at the bottom of its radiator, which held 480 ounces of fluid when it started to leak, and started losing radiator fluid at a constant rate of 4 ounces per minute. Suppose that the radiator continued to leak at this constant rate and that the truck, traveling at 35 miles per hour, could continue traveling at this rate until its radiator was completely empty. In how many *miles* would the radiator be empty?

- F. 13.7
- G. 17.5
- H. 35.0
- J. 70.0
- K. 120.0

35. Each of 3 lines crosses the other 2 lines, as shown below. Which of the following relationships, involving angle measures (in degrees), *must* be true?



- I. $m\angle 4 + m\angle 5 + m\angle 10 = 180^\circ$
- II. $m\angle 2 + m\angle 7 + m\angle 12 = 180^\circ$
- III. $m\angle 2 + m\angle 7 + m\angle 10 = 180^\circ$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II, and III

36. How many ordered pairs (x, y) of real numbers will satisfy the equation $2x - 5y = 6$?

- F. 0
- G. 1
- H. 2
- J. 3
- K. Infinitely many

37. If Mark works steadily he can complete a task in x hours. What portion of the task remains if he works steadily for y hours, where y is any value less than x ?

- A. $\frac{(x+y)}{x}$
- B. $\frac{(y-x)}{x}$
- C. $\frac{(x-y)}{(x+y)}$
- D. $\frac{(x-y)}{y}$
- E. $\frac{(x-y)}{x}$

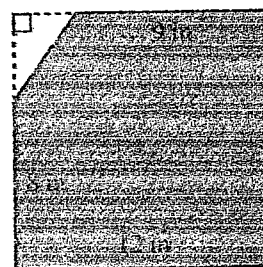
38. If $a^2 - b^2 = 81$ and $a - b = 9$, then $a = ?$

- F. 12
- G. 9
- H. 3
- J. -3
- K. -9

39. For $y \neq 0$, $\frac{y^8}{y^2}$ is equivalent to:

- A. 1
- B. 4
- C. y^3
- D. y^4
- E. y^6

40. The polygon below was a square with 12-inch sides before a triangle was cut off. What is the perimeter, in inches, of this polygon?



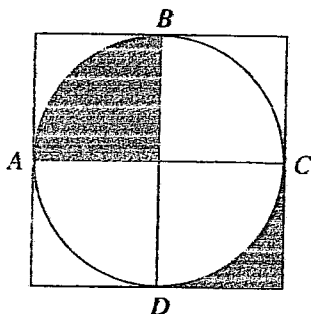
- F. 41
- G. 43
- H. 46
- J. 48
- K. 53

41. A circle in the standard (x, y) coordinate plane has center $(2, -3)$ and radius 4 units. Which of the following equations represents this circle?

- A. $(x-2)^2 + (y+3)^2 = 4$
- B. $(x+2)^2 - (y-3)^2 = 4$
- C. $(x+2)^2 + (y-3)^2 = 4$
- D. $(x-2)^2 + (y+3)^2 = 16$
- E. $(x+2)^2 - (y-3)^2 = 16$



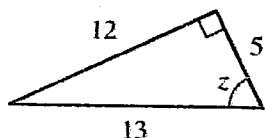
42. In the figure below, a square is circumscribed about a circle with a 30-inch diameter. Points A , B , C , and D are the midpoints of the square's sides. What is the total area, in square inches, of the shaded regions?



- F. 450
G. 225
H. 177
J. 94
K. 47

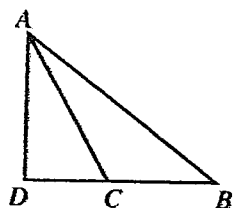
43. For the triangle shown below, what is the value of $\tan z$?

- A. $\frac{5}{12}$
B. $\frac{12}{5}$
C. $\frac{5}{13}$
D. $\frac{13}{5}$
E. $\frac{12}{13}$



44. In $\triangle ABD$ below, points D , C , and B are collinear, $\overline{AD}$ is perpendicular to $\overline{DB}$, and $\overline{AC}$ bisects $\angle DAB$. If the measure of $\angle CBA$ is 40° , what is the measure of $\angle ACB$?

- F. 115°
G. 112.5°
H. 110°
J. 107.5°
K. 105°



45. You have enough material to build a fence 40 meters long. If you use it all to enclose a square region, how many square meters will you enclose?

- A. 160
B. 100
C. 80
D. 40
E. 20

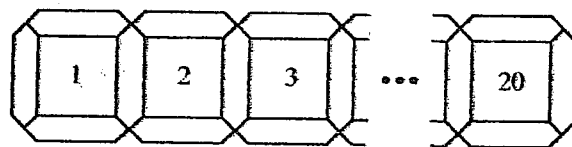
46. For what nonzero whole number k does the quadratic equation $x^2 + k^2x + 2k = 0$ have exactly 1 real solution for x ?

- F. -4
G. -2
H. 2
J. 4
K. 8

47. For all $x > 3$, $\frac{3x - x^2}{x^2 + 3x - 18} = ?$

- A. $\frac{-x}{x+6}$
B. $\frac{x}{x-6}$
C. $\frac{1}{x+6}$
D. $-\frac{1}{18}$
E. $\frac{1}{18}$

48. The pattern shown in abbreviated form below is composed of squares that are arranged horizontally and surrounded by 4 hexagons. All the squares are congruent, and all the hexagons are congruent. How many of these congruent hexagons will there be if the pattern is repeated until there are 20 squares?



- F. 44
G. 61
H. 70
J. 79
K. 80

49. If the circumference of a circle is $\frac{5}{2}\pi$ inches, how many inches long is its radius?

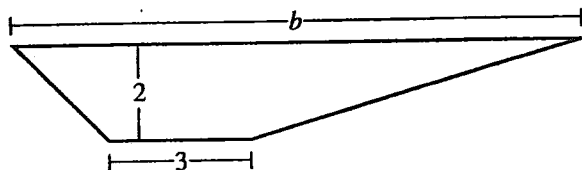
- A. $\frac{2}{5}$
B. $\frac{4}{5}$
C. $\frac{5}{4}$
D. $\frac{\sqrt{5}}{2}$
E. $\sqrt{\frac{5}{2}}$

50. If the function f satisfies the equation $f(x + y) = f(x) + f(y)$ for every pair of real numbers x and y , what is(are) the possible value(s) of $f(0)$?

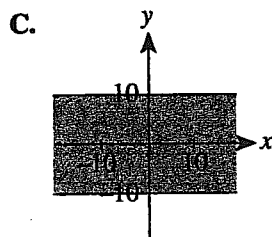
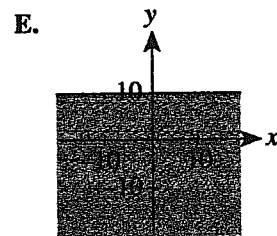
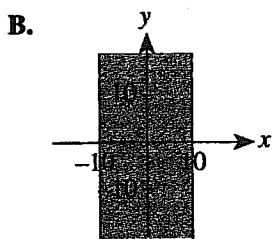
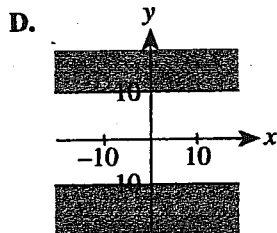
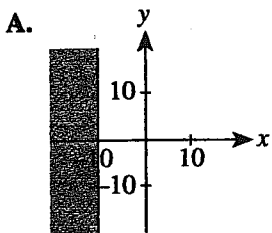
- F. Any real number
G. Any positive real number
H. 0 and 1 only
J. 1 only
K. 0 only



51. The area of the trapezoid below is 16 square inches, the altitude is 2 inches, and the length of one base is 3 inches. What is the length, b , of the other base, in inches?



- A. $2\frac{2}{3}$
 B. 5
 C. 11
 D. 13
 E. 15
52. This year, 75% of the graduating class of Harriet Tubman High School had taken at least 8 math courses. Of the remaining class members, 60% had taken 6 or 7 math courses. What percent of the graduating class had taken fewer than 6 math courses?
- F. 0%
 G. 10%
 H. 15%
 J. 30%
 K. 45%
53. Which of the following shaded regions is the graph in the standard (x,y) coordinate plane of the points that satisfy the inequality $|y| \leq 10$?



54. A standard sheet of paper ($8\frac{1}{2}$ inches wide by 11 inches long) is rolled up so that the 2 shorter edges just meet, forming a circular tube (cylinder) $8\frac{1}{2}$ inches high. How many inches in circumference is the tube?

- F. 11
 G. $\frac{11}{2\pi}$
 H. 22π
 J. $\frac{289}{4}\pi$
 K. 121π

55. A baseball team played its first 20 games and won 12 of them. Then, the team went on a losing streak and lost its next 4 games. How many consecutive additional victories does the baseball team need in order to bring its winning percentage back to at least what it was just before this 4-game losing streak?

- A. 2
 B. 4
 C. 6
 D. 8
 E. 12

56. Lines p and q intersect at point $(1,3)$ in the standard (x,y) coordinate plane. Lines p and r intersect at $(2,5)$. Which of the following is an equation for line p ?

- F. $y = 2x + 1$
 G. $y = 2x + 2$
 H. $y = 2x + 3$
 J. $y = 2x + 5$
 K. Cannot be determined from the given information

57. If $\sin \alpha = \frac{3}{4}$, and α is the measure of an acute angle, then $\cos \alpha = ?$
 (Note: An acute angle has a degree measure from 0° to 90° .)

- A. $\frac{1}{4}$
 B. $\frac{4}{3}$
 C. $\frac{\sqrt{3}}{2}$
 D. $\frac{\sqrt{7}}{4}$

- E. Cannot be determined from the given information



58. The radio station WEST is erecting a new transmitting tower that is 280 feet tall. A support wire will be attached to the ground at point A and to the tower 250 feet up at point B , as shown below. The wire must be at least as long as $\overline{AB}$. Which of the following expresses the length of $\overline{AB}$, in feet?

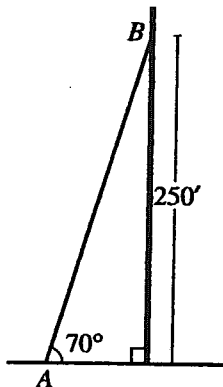
F. $250 \cos 70^\circ$

G. $250 \sin 70^\circ$

H. $250 \tan 70^\circ$

J. $\frac{250}{\cos 70^\circ}$

K. $\frac{250}{\sin 70^\circ}$



59. In an arithmetic series, the terms of the series are equally spread out. For example, in $1 + 5 + 9 + 13 + 17$, consecutive terms are 4 apart. If the first term in an arithmetic series is 3, the last term is 136, and the sum is 1,390, what are the first 3 terms?

A. 3, 10, 17

B. 3, 23, 43

C. 3, $36\frac{1}{3}$, 70

D. 3, $69\frac{1}{2}$, 136

E. 3, 139, 1,251

60. Right triangle $\triangle ABC$ has angle measures α , β , and γ degrees and side lengths a , b , and c inches, as illustrated below. Which of the following is true about the value of the product $\tan \beta \tan \gamma$?

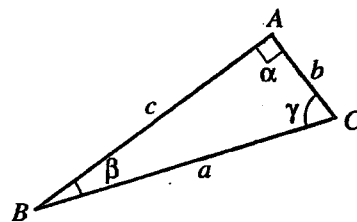
F. The value is 1.

G. The value is $\frac{b^2}{c^2}$.

H. The value is $\frac{b^2 c^2}{a^4}$.

J. The value is undefined.

K. The value cannot be determined from the given information.



END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the short story "Ghirlandaio" by Francine Prose (©1993 by Francine Prose).

On the morning of the trip to the art museum I woke up shaking with fever. I still remember staring down into my dresser drawer, wondering how many sweaters I could get away with wearing. I must have
5 put on three or four, but nothing felt warm. At breakfast, I shivered and tried to hide it. How strange that my parents didn't notice; normally, one sniffle and they were feeling my forehead. But sometime during the night we must have entered that world of mischance
10 that parents so fear, with its history of catastrophes occurring in eye blinks when parental vigilance lapsed.

Briefly I wondered if maybe I did have polio, as my mother so dreaded, but I was still a child, and didn't know what was worth fearing; children rarely fear air-
15 planes but, almost always, the dark. The prospect of missing the trip scared me far more than polio. Besides, I already knew that first principle of everyday magic: once you say something, give it a name, then, only then, can it happen. So I kept quiet and shivered and
20 wrapped my hands around my cocoa cup and everything around me slipped in and out of focus.

This is how I recall that day—at moments the edges of things would be painfully sharp; then they would blur and turn wavy. Kissing my parents goodbye,
25 I was so confused I imagined my father would be interested to hear that the world looked to me like an El Greco painting. But just in time I caught myself and climbed onto the steamed-up bus.

Our classroom was in chaos, but through it all rang
30 Miss Haley's strained voice, yelling, "Hang on to your coats," which struck me as the most deeply kind, the most thoughtful thing she'd ever said. There was one moment, as we lined up to leave, when I knew I was in danger, that I should tell someone and go home. But
35 then I felt someone bump into me, and even through all those sweaters, I knew who it was. Kenny was right behind me in line, and as we pushed toward the narrow bus door, he whispered, "Can we still go see it?" It took me a while to think what he meant, though for days it
40 was all I had thought of.

What he meant was the Ghirlandaio painting, which he'd heard about from me. It had required astonishing bravery to approach him in the schoolyard, to speak to him for so long, but that was minor compared
45 with the courage it took to mention the unmentionable—that is, Miss Haley's nose. I don't recall how I'd phrased it, how precisely I'd made it clear that there existed a work of art with a nose like our sixth-grade teacher's. It had left us both feeling quite short of
50 breath, as if we'd been running and had gotten our second wind and were capable of anything. And in that light-headed state I offered to take him to see it. It would be easy, I said—I knew the museum so well we could sneak off and get back before anyone noticed.

Yet now the idea of walking even the shortest distance exhausted me, and my plan (which I'd never expected him to agree to) seemed to demand impossible stamina—though less than it would have taken to shake
55 my head no. I told him to be on the lookout for the right moment, and my voice dopplered back at me through an echo chamber of fever.

At the museum, a guard instructed us to throw our coats in a rolling canvas bin. And this is my clearest memory from that day—the panic I felt as my coat dis-
65 appeared, how it looked to me like someone jumping, vanishing into a sea of coats. Suddenly I was so cold I felt I had to keep moving, and I caught Kenny's eye and we edged toward the back of the crowd, and dimly I heard my fever-voice telling him: Follow me.

Not even running helped. I just got colder, wobbly, and unsure; of course we got lost and crisscrossed the damp medieval hall, where the shadows climbed the chill stone walls, pretending to be door-
70 ways that vanished when we got close. At last we found the staircase, the right gallery, the Ghirlandaio. And I gloried in the particular pride of having done what I'd boasted I could.

Kenny stared at the painting. Then very softly he said, "Wow. Disgusto."

"Disgusto" was the word, all right. And yet I felt strangely hurt, protective of Ghirlandaio's old man, as if he and his grandson were relatives of mine and
80 Kenny had passed judgment on my family, on my life,

85 on those afternoons when I stood here with my father pretending that this was something compelling and beautiful and not what it was: disgusto.

1. As it is used in the passage (line 58), the word *stamina* most nearly means:
 - A. strength.
 - B. silence.
 - C. boldness.
 - D. quickness.
2. The narrator states that she was most afraid of:
 - F. contracting polio.
 - G. missing the trip.
 - H. being in the dark.
 - J. naming her illness.
3. The passage suggests that the narrator was first introduced to the Ghirlandaio painting:
 - A. in a classroom art lesson given by Miss Haley.
 - B. in an art book presented to her by her parents.
 - C. on a previous visit to the art museum.
 - D. on the museum trip with Kenny and her classmates.
4. The narrator states that she dimly heard her fever-voice tell Kenny, "Follow me" (lines 68–69). This description suggests that the narrator:
 - F. was speaking very quietly to avoid being overheard by the museum guard.
 - G. was experiencing difficulty in hearing because of her classmates' excited conversation.
 - H. felt that her illness had taken away her control of her speech.
 - J. believed that her "fever-voice" would more readily persuade Kenny.
5. The narrator considers her parents' behavior, as it is described in lines 6–11, to be:
 - A. habitually indifferent.
 - B. unusually lenient.
 - C. particularly strict.
 - D. unusually inattentive.
6. Kenny's reaction to the Ghirlandaio painting can best be described as:
 - F. threatened; he feels the painting is ominous.
 - G. awestruck; he reacts strongly to the painting.
 - H. rational; he wants his comments on the painting to make sense.
 - J. discreet; he doesn't want the teacher to hear him.
7. The narrator's "plan," mentioned in line 56, depends most critically upon:
 - A. the distraction caused by her classmates.
 - B. the convenient location of the painting.
 - C. Kenny's proficiency as a student.
 - D. her familiarity with the art museum.
8. Discussing the resemblances between the old man in the Ghirlandaio painting and their teacher leaves Kenny and the narrator feeling:
 - F. exhilarated.
 - G. exhausted.
 - H. lethargic.
 - J. irritated.
9. The description in lines 80–86 suggests that Kenny's comment causes the narrator to:
 - A. become defensive about the people in the painting.
 - B. pass judgment on her family and her life.
 - C. romanticize the subject matter of the painting.
 - D. confirm her own doubts about Ghirlandaio's talent.
10. According to her account of the story, the narrator's biggest challenge was to:
 - F. convince Kenny to leave their other classmates while at the museum.
 - G. broach the subject of Miss Haley's nose while speaking to Kenny.
 - H. behave as though she were not ill once she finally saw the painting.
 - J. work up enough courage to approach Kenny in the schoolyard for the first time.

Passage II

SOCIAL SCIENCE: This passage is adapted from testimony given by Renée Askins at a congressional hearing on the reintroduction of wolves to Yellowstone National Park (©1995 by the Harper's Magazine Foundation).

If I were a rancher I probably would not want wolves returned to the West. If I faced the conditions that ranchers face in the West—falling stock prices, rising taxes, prolonged drought, and a nation that is eating less beef and wearing more synthetics—I would not want to add wolves to my woes. I would want to blame something, to fight something.

The wolf is an ideal target: it is tangible, it is blamable, and it is real. Or is it? When ranchers talk about wolves they say, "You know, it's not the wolves we're worried about, it's what the wolves represent; it's not what they'll do, it's what they mean." Wolves mean changes. Wolves mean challenges to the old ways of doing things. Wolves mean loss of control.

Ranchers deserve our compassion and our concern. Whether the threat of wolves is imagined or actual, the ranchers' fear and anger are real.

Ranchers claim that wolves will devastate the livestock industry in the West. Yet all the studies show that wolves kill far less than 1 percent of the livestock available to them. According to the *Bozeman Chronicle*, even if federal specialists have wildly underestimated the number of cows and sheep that wolves would kill in the Yellowstone and central Idaho areas, the actual total would be much smaller than the number that die each year in the state of Montana alone because of storms, dogs, and accidents.

In effect, the livestock industry has successfully transferred to the general public one of its most basic operational costs: prevention of predator losses. If you raise Christmas trees, part of the cost and risk of doing business is losing a few trees to gypsy moths and ice storms; inherent in the cost of ranching, particularly on public lands, should be the cost and risk of losing livestock to predators. Instead, every year 36 million tax dollars go to kill native predators on our public lands so that private industry can make a profit.

It is important to remember that wolves are missing from the Yellowstone region only because we eliminated them. They did not vanish from the area in response to loss of prey or lack of habitat; they did not die out as a result of disease or natural catastrophe. We systematically, intentionally, consciously killed every wolf we could find.

Opponents of wolf reintroduction assume that because there are no wolves, there should be no wolves. They have promoted the idea that the return of wolves is somehow radical or extreme, some sort of environmental luxury, some romantic nonsense that only urbanites and rich Easterners advocate at the expense of the poor, beleaguered Western livestock industry. (In

fact, surveys show that Westerners support the reintroduction.) The industry's cry of economic loss has eclipsed the costs to the general public of not having wolves. In the West we now live in a "wolf-free" environment. Or is it "wolf-deprived"? Who has gained and who has lost? How do we assign a value to the importance of a predator in the ecosystem? How do we determine the cost of removing one note from a Mozart symphony, one sentence from a Tolstoy novel, or one brush stroke from a Rembrandt? Having wolves in Yellowstone is not a luxury but a right. We should not have to pay for clean air or water, nor should we believe that they are somehow a luxury. Similarly, we have a right to a full complement of wildlife on our public lands.

Emotions, not facts, have controlled the wolf debate. Wolves have never been just wolves: the wolf is the devil's keeper, the slayer of innocent children, the nurturer of abandoned babies, the sacred hunter, the ghostly creature of myth and legend. In short, wolves are symbolic; Yellowstone is symbolic; restoring wolves to Yellowstone is a deeply and profoundly symbolic act.

We are a culture of symbols. It is not surprising that ranchers and environmentalists use the symbolic force of wolves to debate painful changes. We use symbols to help us order and make sense of an increasingly complex world. The Yellowstone wolf-recovery debate is fundamentally an expression of a culture in transition; it is the struggle that accompanies old assumptions clashing against the new. The story of this conflict is the story of how we view ourselves in relation to animals, whether we can replace the assumption of "dominion" that has been so destructive to us and the natural world with a worldview that recognizes that we live in a state of reciprocity with the birds and the beasts—that we are not only the product of nature but also part of it.

11. As she is revealed in the passage, Askins can most reasonably be characterized as:

- A. intolerant of ranchers' views regarding wolf reintroduction, since they differ from her own views so completely.
- B. believing that opponents of wolf reintroduction are depriving all Americans of their right to a full complement of wildlife on public lands.
- C. supportive of the livestock industry's use of tax dollars to compensate for predator losses experienced by ranchers using public lands.
- D. surprised that the wolf is the target of so much dislike and at the center of so much controversy, since she herself admires wolves.

12. As it is used in line 65, the word *complement* most nearly means:
- F. flattering remark.
 - G. equal cost.
 - H. selection.
 - J. safe number.
13. It can reasonably be inferred that Askins would most likely agree with which of the following statements?
- A. People's historic domination of certain species of animals has been both necessary and beneficial to humans and the natural world.
 - B. The idea of reintroducing wolves to an area from which they have been absent for twenty years or more is a luxury promoted mainly by wealthy residents of the East.
 - C. Ranchers and environmentalists alike have primarily used well-documented facts to support their arguments in the wolf reintroduction debate.
 - D. People are solely responsible for the disappearance of wolves from Yellowstone, so it is appropriate that people should attempt to reintroduce wolves to that environment.
14. According to Askins, ranchers are less worried about wolves than they are worried about:
- F. losing control over the environment in which they raise livestock.
 - G. maintaining the luxury of clean air and clean water on public lands.
 - H. their tendency to view themselves as both a product of nature and a part of it.
 - J. comparing the number of livestock killed by storms, dogs, and accidents.
15. It can most reasonably be inferred that Askins asks the rhetorical question (a question to which she expects no answer) "How do we determine the cost of removing one note from a Mozart symphony, one sentence from a Tolstoy novel, or one brush stroke from a Rembrandt?" (lines 58–61) in order to:
- A. prove that the cost of reintroducing wolves to the Yellowstone environment would be as high as the cost of making the changes she lists in the question.
 - B. demonstrate her belief that the debate about reintroducing wolves to the Yellowstone environment cannot be assessed in purely economic terms.
 - C. educate readers about the market value of symphonies, works of literature, and paintings, which is similar to the cost of returning wolves to the Yellowstone environment.
 - D. test readers' knowledge about the economic impact of the changes she lists, since she believes these costs can be determined.
16. Which of the following statements best describes Askins's method of and purpose for addressing her subject?
- F. She has relayed a series of personal anecdotes and memories in an attempt to persuade members of Congress to agree with her.
 - G. She has presented her personal opinion supported by factual information to try to persuade members of Congress to support wolf reintroduction.
 - H. She has constructed an argument based on emotion with no supporting facts to members of Congress in the hope that they will sympathize with her viewpoint.
 - J. She has presented a series of researched facts designed to show members of Congress the opposing viewpoints of the wolf reintroduction debate.
17. It is most reasonable to infer that when Askins claims that "restoring wolves to Yellowstone is a deeply and profoundly symbolic act" (lines 72–74) she means that reintroducing:
- A. wolves is an act that can be important to people in theory only, not in fact.
 - B. any species to an area is a symbol of humans' moral obligation to maintain previously established methods of controlling animal populations.
 - C. any species to an area is one way humans can acknowledge their rightful place as rulers of the natural world.
 - D. wolves represents an acknowledgment by humans of the importance of including a species formerly considered an enemy.
18. As it is used in line 54, the word *eclipsed* most nearly means:
- F. highlighted.
 - G. echoed.
 - H. obscured.
 - J. exaggerated.
19. It can most reasonably be inferred that Askins compares the cost of losing Christmas trees to moths and storms to the cost of losing livestock to predators (lines 30–35) because she believes that:
- A. they are both examples of costs ranchers should not have to bear.
 - B. they are both examples of costs the public should bear.
 - C. ranchers should absorb the cost of losing livestock to predators.
 - D. the public should absorb the cost of losing livestock to predators.

20. According to Askins, ranchers' difficulties include their being affected by which of the following trends?
- F. The public's growing support for spending tax dollars to kill native predators on public lands
 - G. Federal specialists' inaccurate estimates of the number of cows and sheep that wolves would kill in a given area
 - H. A desire on the part of the livestock industry for a "wolf-free" environment
 - J. People's growing tendency to eat less beef and wear fewer leather clothing items

Passage III

HUMANITIES: This passage is adapted from Indira Ganesan's essay "Resisting My Family History," which appeared in *Glamour* magazine (©1994 by The Condé Nast Publications Inc.).

In June 1978, in suburban Nanuet, New York, home to a famous mall, I wore a *sari* [a garment worn by Indian and Pakistani women] to my high school graduation. The next day it was raining, and my family and I set off for India. Watch out for the cows, wrote my favorite English teacher, tongue-in-cheek, in my yearbook. I'd keep a wary eye out, I thought, packing a dozen cassettes on which I'd recorded my favorite music. My friends stood in the rain with a banner proclaiming *Goodbye, Indira!* I was a heroine, a star for the moment. I was leaving the country.

Though I'd been born there, I didn't want to go to India. What I wanted was to knock on a Broadway producer's door and say, "I'm brown, I'm talented, let me write you a play." My parents, however, believed I needed to embrace my Indian past. I wanted only to escape it, as I wanted to escape anything that spoke to me of tradition or old-fashioned ideas. I was too cool for India.

I remember the heat in Bombay's airport, so thick it was sliceable; the crush of people who wanted to help after our car developed engine trouble on our way home; the glare of the outdoors; the cool dark of the interiors. My uncle's family welcomed my mother and me into their home in Madras, a city on the southeastern part of the subcontinent, famed for its music festivals and its beach. In Madras I enrolled in a Catholic women's college. Nuns were the teachers, the English language was the norm.

In high school I had edited an underground newspaper, bought my first copy of *The Village Voice*, read *The New York Times* regularly. I believed I was a feminist. In India I was unsure of my role. Above all, I was deeply worried that I'd be married off, that I'd be forced to become a housewife, horror of horrors, and would lose my freedom.

What I discovered in India: people who looked like me. Girls who befriended me instantly. Girls who told me the truth at once. I was an American, and how

everyone knew that—for they all did—escaped me. Wasn't I as brown as they? Or was I giving off an American aura, wearing Wrangler jeans and a T-shirt, speaking hesitant Tamil?

I attended a wedding. I watched some of the funeral preparations following the death of my great-uncle. I climbed 500 steps to reach a Jain temple where a priest gave me a blessing that translated roughly as "You will have seven years of good luck followed by seven years more of the same." During a ten-day tour of famous temples, I saw a snake charmer in a parking lot and visited an entire city of priests and ascetics. They let us into a temple's sanctum sanctorum, where we saw the God image in all its splendor.

In India I had the unswerving consideration of my relatives, 25 of whom I met in my first six months. I remember our meals together, and the preparations: the pile of freshly shredded coconut—white, flaky, fragrant with milk; the way sweet dough for *jellabies* would be dropped in hot oil and bob up to perfection. In America I picked at pizza and baked ziti on the school lunch menu; in India I feasted. There were scores of delicious meals, piles of snacks in tins, water always available in an earthen vessel in the kitchen.

And still I felt I was missing out on a superlative year in America, and I was determined to dislike India. I dragged my aunt to see a Woody Allen movie and felt it superior to Indian films, even though they managed to reduce me to tears.

In all of this, I, the *yanqui*, was a source of amusement to the family. They bent over backward to please me that year, and I finally admitted to enjoying myself. I can still recall the din of the streets as I rushed to college in the mornings—bicycles, rickshaws, buses, pedestrians, bikes and yes, even a bull here and there.

Now a teacher myself in San Diego, I have just come back from giving a class on "the travel essay." One of my students suggested that the writer we were studying had formed her opinion of the country she was visiting before traveling there. In a sense, I thought, that is what I did with my year in India. I knew before going that I would like the temples and the food and the embrace of my relatives; I just didn't think it was a place for me, a newly graduated high school senior who dreamed of travel. How wrong I was. It was my year abroad, a high-seas adventure from which I would draw for years to come. In India I took my dreams of becoming a someone and began to be a someone. A someone connected to a history, to a family, to a distinct geography. A someone who had traveled after all.

21. The passage suggests that one quality the narrator appreciated about the girls she met in India was their:

- A. feminist viewpoint.
- B. taste in music.
- C. adventurousness.
- D. openness.

22. The passage suggests that the narrator dragged her aunt to a Woody Allen movie because:
- F. the narrator refused to attend Indian movies.
 - G. her aunt did not know who Woody Allen was.
 - H. the narrator was determined to cling to the American culture she left behind.
 - J. the narrator wanted to prove that Indian values were just as important as American values.
23. The passage states that instead of traveling to India after graduating from high school, the narrator had hoped to:
- A. attend college in New York.
 - B. travel in the U.S. with her friends.
 - C. become a playwright.
 - D. act in a Broadway play.
24. The narrator's claim that "I had the unswerving consideration of my relatives" (lines 54–55) suggests that her relatives:
- F. treated her as though she were still a child.
 - G. treated her with thoughtful concern.
 - H. were often afraid of hurting her feelings.
 - J. were stubborn people.
25. Information in the sixth paragraph (lines 44–53) supports the narrator's claim that:
- A. her trip to India was an adventure.
 - B. she was missing out on a superlative year in the U.S.
 - C. everyone in India knew she was American.
 - D. she was determined to dislike India.
26. The fact that the narrator wore a *sari* to her high school graduation seems to contradict her claim that:
- F. she was worried about being married off in India.
 - G. she spoke only in hesitant Tamil.
 - H. she wanted to escape her Indian past.
 - J. her high school classmates thought she was a heroine.
27. In the seventh paragraph (lines 54–63), the narrator offers details about Indian food most likely to:
- A. present one of the more appealing aspects about life in India.
 - B. illustrate that she appreciates good restaurants.
 - C. illustrate that she is willing to learn cooking skills.
 - D. support her claim that the people of India are very generous.
28. The last paragraph suggests that one of the narrator's students from her travel-writing class caused the narrator to:
- F. regret going to India instead of a more interesting place.
 - G. reconsider the significance of her own journey to India.
 - H. view her Indian relatives in a new way.
 - J. feel guilty about her aloofness when she was in India.
29. The passage states that the narrator's relatives in India viewed her as being:
- A. spoiled.
 - B. condescending.
 - C. intelligent.
 - D. entertaining.
30. The first two paragraphs of the passage establish all the following facts about the narrator EXCEPT that she:
- F. was well-liked by her high school friends.
 - G. anticipated a disappointing time during her trip to India.
 - H. enjoyed spending time in a suburban shopping mall.
 - J. still observed some Indian cultural traditions.

Passage IV

NATURAL SCIENCE: This passage is adapted from *Sarapiqui Chronicle* by naturalist Allen M. Young (©1991 by the Smithsonian Institution).

Converting dying or dead tissues into nutrients is what binds the creatures of the tropical rain forest into a functioning unit. Thus the giant trees, what I have come to love and admire about Sarapiqui [rain forest] when I stop at a ridge above the forest canopy, are supported precariously upon a thin, fragile tissue of microbes and organic matter, matter that is turned over, transformed by millipedes, sowbugs, ants, and millions of other tiny creatures. Without these hidden, largely unseen assemblages of life, the giants would be no more. But this exquisite association between big trees and the Earth's tiniest creatures has structured and guided the development of Central America's and South America's rain forests for millions of years. With the fixation of energy from the sun through photosynthesis to make living plant tissues, and the continual absence of a cold temperature winter season, the tropical rain forest pulses with energy, most of it tied up in the bodies of its living creatures, corpuscles of nutrients feeding into one another.

What a fascinating, elegant circle of life *Morpho* [butterflies] symbolizes in this regard. Plants die in the rain forest, including the woody vines *Morpho* caterpillars feed upon and are evolutionarily specialized to exploit, and saprophytic fungi and bacteria attack the dead plant material. In the process of breaking down dead plant material to feed themselves, the fungi metabolize substances that become attractive to *Morpho*. The eventual breakdown of the dead plant material provides the rain forest with the fertilizer it needs in order to survive, including the woody vines and other legumes fed upon by *Morpho* caterpillars. *Morpho* also gets nutrients from the decay organisms as well.

Morpho symbolize a great deal about the workings of a tropical rain forest. When mushrooms break down the dying and dead tissues of a log or tree trunk, these saprophytic organisms are releasing valuable nutrients into the rain forest, to be fed upon by living plants. So too when the very large grub of a *Megasoma* or *Dynastes* scarab beetle, what the Costa Ricans call the *cornizuelo*, ingests the rotting wood of a tree stump, it is also unlocking essential nutrients to nurture the rain forest. And when birds peck holes in the trunks of trees, exposing sap that soon becomes encrusted with fermenting mold. Such mold, like that of rotting fruit in the forest's canopy or on its floor, converts nature's most ubiquitous fuel molecule, sugar, into the structural building blocks of life, proteins and other substances. *Morpho*'s own existence, together with that of legions of other insects and other arthropods that disperse bacteria and spores, ensures this rain forest will nourish *Mucuna* and other legumes that feed this butterfly's curious red and yellow caterpillars.

But this intimate biological partnership between microbes and big trees in the tropical rain forest is being broken apart by deforestation. And as I write, this situation has tragic implications for *Morpho*, creatures whose own lives are also a cognitive piece of the living tapestry of Sarapiqui's forests. It takes about 150 *Mucuna* leaves to make one mature caterpillar of *Morpho peleides*, or a wet weight of about 20 grams of plant flesh to produce one butterfly. Typically a clump of *Mucuna* or *Machaerium* occupies little more than a few square meters of habitat floor space, an area not even the girth of an average canopy tree of the tropical rain forest. For one *Morpho* or several individuals growing up in the same vine patch, not much forest space is required. But the food resources of morphoes, their caterpillar food plants and the juices of decay for the adults, are spread out through the rain forest. Thus the existence of a single *Morpho* is spread out through the forest as well. *Morpho* cannot survive in just one small place: yet it takes about ten acres of tropical pasture to fatten a Brahma steer for slaughter.

How many more mornings when the rain forest bathes in tropical sunshine will there be for me, anyone, to witness the incredible beauty of *Morpho* dancing above the Tirimbina creek? And how much opportunity to see this will there be for much of the floodplain of Sarapiqui? Outside of the La Selva Biological Reserve and the adjoining Braulio Carillo National Park, much of what has been home for *Morpho* is rapidly becoming beef cattle pastures.

31. The last paragraph suggests that the author of the passage is:
 - A. an eminent international naturalist.
 - B. an enemy of the La Selva Biological Reserve.
 - C. concerned about the future of the rain forest.
 - D. an infrequent visitor to Sarapiqui's forests.
32. The passage indicates that the *cornizuelo* eats:
 - F. scarab beetles.
 - G. mushrooms.
 - H. *Morpho*.
 - J. rotting wood.
33. The "assemblages of life" referred to in line 10 are:
 - A. the rain forest's smallest creatures.
 - B. large groups of rain forest mammals.
 - C. *Morpho* caterpillars and butterflies.
 - D. mushrooms and scarab beetles.

34. The main worry expressed in the fourth paragraph (lines 55–75) is that:
- F. *Morpho* are a cognitive portion of a living tapestry.
 - G. deforestation is damaging a delicate natural balance.
 - H. deforestation is ravaging several beautiful countries.
 - J. Sarapiquí's forests will have fewer butterflies.
35. The author's attitude toward the study of the life cycle of *Morpho* is best characterized as one of:
- A. scientific detachment.
 - B. excited interest.
 - C. scholarly indifference.
 - D. dispassionate observation.
36. The passage identifies *Mucuna* as a:
- F. caterpillar.
 - G. butterfly.
 - H. mushroom.
 - J. legume.
37. Lines 35–36 suggest about *Morpho* that they are:
- A. an unusual part of a living tapestry.
 - B. an exceptionally beautiful yet rare butterfly.
 - C. a representative rain forest species.
 - D. a saprophytic organism of rare elegance.
38. The passage suggests that *Morpho* might also be correctly identified as:
- F. a legume.
 - G. an ubiquitous fuel.
 - H. a *cornizuelo*.
 - J. an arthropod.
39. The passage states that *Morpho* grow up in:
- A. the forest canopy.
 - B. vine patches.
 - C. tree trunks.
 - D. rotten logs.
40. The passage claims that one of the biological functions of arthropods is to:
- F. spread spores throughout the rain forest.
 - G. produce the fertilizer trees rely on.
 - H. facilitate the increase in rain forest habitat.
 - J. break down dead plant material for the *Mucuna*.

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END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.
DO NOT RETURN TO A PREVIOUS TEST.

SCIENCE REASONING TEST*35 Minutes—40 Questions*

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage 1

Herbicides are used to control the growth of weeds. An herbicide that may be used safely with one crop species may damage another crop if the latter crop is planted in soil containing residual amounts of the herbicide from an earlier application. Two experiments were performed to study this effect.

Experiment 1

A botanist filled 90 pots with Soil Type 1. No herbicide was added to the soil in 10 pots. The other pots were divided into groups of 10 and the soil in each group was treated with 10, 20, 50, or 100 ppm of either Herbicide A or B. All other factors were held constant. Ten seeds of a corn hybrid were planted in each pot. After 40 days, the plants were uprooted, oven-dried, and weighed. The results are shown in Table 1.

Table 1		
Herbicide dose (ppm)	Average mass of plants (g)	
	Herbicide A	Herbicide B
10	14.1	15.6
20	12.4	13.7
50	9.3	12.1
100	5.5	9.3
Note: Average plant mass in untreated soil was 16.0 g.		

Experiment 2

Experiment 1 was repeated with 90 pots of Soil Type 1 and 90 pots of Soil Type 2. The same herbicide doses and corn hybrid were used. All other factors were held constant. After 40 days, the heights of the plants were measured. The results are shown in Table 2.

Table 2				
Herbicide dose (ppm)	Average height of plants (cm)			
	Soil Type 1		Soil Type 2	
	Herbicide A	Herbicide B	Herbicide A	Herbicide B
10	46.3	49.0	50.3	52.5
20	42.0	47.0	44.4	47.0
50	34.1	39.4	40.6	42.3
100	19.6	22.7	30.9	36.4
Note: Average plant height in untreated Soil Type 1 was 50.6 cm; average plant height in untreated Soil Type 2 was 52.7 cm.				

Information on the two soil types used is given in Table 3.

Table 3			
Soil Type	pH	Organic matter (%)	Clay (%)
1	6.9	5.0	16.3
2	6.2	9.5	7.9



1. The results of Experiment 2 indicate that, at every herbicide dose, average plant height was *lowest* under which of the following conditions?
 - A. Herbicide A and Soil Type 1
 - B. Herbicide B and Soil Type 1
 - C. Herbicide A and Soil Type 2
 - D. Herbicide B and Soil Type 2
2. Which of the following sets of plants served as the control in Experiment 1?
 - F. Plants grown in untreated soil
 - G. Plants grown in soil treated with 10 ppm of Herbicide A
 - H. Plants grown in soil treated with 10 ppm of Herbicide B
 - J. Plants grown in soil treated with 100 ppm of Herbicide A
3. Which of the following best explains why the herbicides were applied to the soil instead of directly onto the corn plants?
 - A. Corn plants are not affected when herbicides are applied directly on them.
 - B. Corn plants usually die immediately upon application of herbicides.
 - C. The experiments were testing how herbicides present in the soil affect corn growth.
 - D. The experiments were testing how soil pH affects corn growth.
4. Assume that there is a direct correlation between plant height and plant mass. If Experiment 1 were repeated using Soil Type 2, one would predict that the average plant mass would be *lowest* under which of the following conditions?
 - F. Herbicide A at 20 ppm
 - G. Herbicide B at 50 ppm
 - H. Herbicide A at 100 ppm
 - J. Herbicide B at 100 ppm
5. Assume that a second corn hybrid was grown in soil treated with varying doses of a third herbicide (Herbicide C). Based on the results of the experiments, what prediction, if any, about the effect of Herbicide C on the growth of this second corn hybrid can be made?
 - A. Herbicide C would have no effect on the growth of these plants.
 - B. Herbicide C would interfere with plant growth, but only at doses above 50 ppm.
 - C. Herbicide C would interfere with plant growth at low doses, but have no effect at high doses.
 - D. No prediction can be made on the basis of the results.
6. To investigate the impact of soil acidity alone on the effect of the herbicides on corn growth, the botanist should design experiments using soils of:
 - F. varying pH, percent organic matter, and percent clay.
 - G. varying pH and percent organic matter, but identical percent clay.
 - H. varying pH and percent clay, but identical percent organic matter.
 - J. varying pH, but identical percent organic matter and identical percent clay.



Passage II

Two scientists discuss the possibility of accurate short-term earthquake prediction (accurately predicting a quake's strength and location within hours or days).

Scientist 1

Short-term earthquake predictions can be made at the present time by detecting events that occur shortly before a quake. Rocks increase in volume under the great pressures often generated in fault zones. At the same time, numerous cracks open in the rocks. The volume increase can be detected by instruments installed near an active fault. Groundwater is often forced away from the fault by the expansion, raising water levels in nearby wells before a quake. The opening of cracks causes the speed of one type of earthquake wave to decrease compared to the speed that those waves had through the rocks before they cracked. Radon gas (a gas found in rocks in many locations) is released by the cracking, and the released gas can be detected by instruments.

Most quakes are preceded by *foreshocks*. Foreshocks are minor quake vibrations that occur minutes, hours, or days before the large quake. Monitoring *seismographs* (instruments used to detect vibrations along faults) for foreshocks can help predict quakes.

Scientist 2

Short-term earthquake predictions cannot be made at the present time with any accuracy. Long-term predictions (within 10 to 100 years) are much more accurate than short-term predictions. Many of the physical changes that occur in rocks under pressure vary according to the rock type or the variety of rock types that are present in a fault zone. Often, the expected volume increase happens long before the quake occurs, or it happens only moments before a quake. Changes in water levels of wells may be caused by factors unrelated to changes in rocks. The detection of radon is a process that takes weeks to conduct and obtain results. Many fault zones have almost constant vibration going on, and foreshocks are almost impossible to separate from these vibrations. One major barrier to short-term prediction is that it is unlikely we ever will be able to directly study or sample the region 10 km or more below the surface, where quakes occur.

7. Which of the following ideas about short-term earthquake prediction was implied by Scientist 2?

- A. Short-term earthquake prediction will be possible in the near future.
- B. There are too many variables to make short-term earthquake prediction possible.
- C. Short-term earthquake prediction can indicate an earthquake's location.
- D. Short-term earthquake prediction can indicate an earthquake's strength.

8. A scientific article stated that "Humans will probably never be able to drill to a depth of 10 km." Which of the scientists' viewpoints, if any, is(are) in agreement with this statement?

- F. Scientist 1 only
- G. Scientist 2 only
- H. Both Scientists 1 and 2
- J. Neither Scientist 1 nor 2

9. Researchers testing the basements of homes in one area of a city found much higher levels of radon gas than the levels in homes in other parts of the city. Which of the following statements about this finding would both scientists most likely agree with?

- A. That area of the city will have an earthquake within 2 days.
- B. An earthquake will occur within 2 days, but the location cannot be predicted.
- C. This finding about radon gas levels may have nothing to do with earthquakes.
- D. Radon gas level changes are never related to earthquakes.

10. Scientists 1 and 2 would most likely agree with which of the following statements about rocks under pressure?

- F. Rocks under pressure exhibit an increase in volume.
- G. Rocks under pressure produce groundwater.
- H. Rock volume increase is the only explanation for a rise in groundwater level in wells.
- J. All rocks are under the same amount of pressure in fault zones.

11. Which of the following procedures would be the best way for Scientist 1 to help test her hypothesis?

- A. Recording the strength of all earthquakes that have occurred in the past year
- B. Measuring rock volume increases and earthquake wave speeds in a single area not prone to earthquakes
- C. Measuring rock volume increases and earthquake wave speeds in many areas prone to earthquakes
- D. Mapping the location of all earthquakes that have occurred in the past 60 years



12. Suppose a new type of seismograph were invented that is capable of easily distinguishing foreshocks from other vibrations. This ability to easily distinguish foreshocks from other vibrations would:
- F. be inconsistent with the view of Scientist 1 only.
 - G. be inconsistent with the view of Scientist 2 only.
 - H. be consistent with the views of both Scientists 1 and 2.
 - J. have no relevance to either scientist's view.
13. According to Scientist 2, which of the following assumptions about rocks is a major flaw in Scientist 1's view?
- A. All rocks react to pressure in an identical manner.
 - B. The cracking of rocks causes an increase in rock volume.
 - C. Radon gas is present in all types of rock.
 - D. All rocks melt when subjected to pressure.

Passage III

Solar cycles are periodic variations in the Sun's brightness.

Table 1 contains the date of onset for, and duration of, recent solar cycles; the time from a cycle's onset until the Sun reached *solar maximum* (the maximum brightness for that cycle); and the number of sunspots at solar maximum.

Table 1				
Solar cycle number	Date of onset of cycle (month, year)	Duration of cycle (years)	Time from onset of cycle to solar maximum (years)	Number of sunspots at solar maximum
1	December 1878	11.3	5.0	75
2	March 1890	11.8	3.8	88
3	January 1902	11.6	4.0	64
4	August 1913	10.0	4.0	105
5	August 1923	10.1	4.7	78
6	September 1933	10.4	3.6	119
7	February 1944	10.2	3.3	152
8	April 1954	10.5	3.9	201
9	October 1964	11.7	4.1	111
10	June 1976	10.3	3.5	165
11	September 1986	—	2.8	158
Average		10.8	3.9	120

Figure 1 shows the number of sunspots and the Sun's brightness at a particular wavelength for part of one cycle.

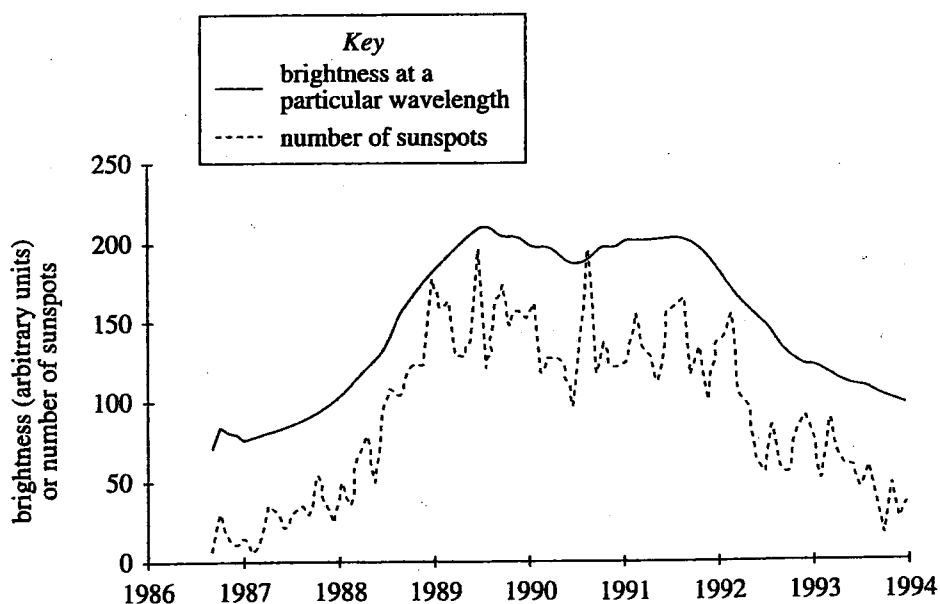


Figure 1

Passage IV

Because amphibian eggs lack a hard outer shell, their DNA can be damaged by exposure to ultraviolet (UV) radiation in sunlight. Table 1 lists the egg-laying behavior of 7 amphibian species and the species' relative ability to repair DNA damage caused by exposure to UV radiation.

Table 1			
Amphibian species	Relative ability to repair DNA damage	Egg-laying behavior	Exposure of eggs to sunlight
A	< 0.1	eggs buried	none
B	< 0.1	eggs laid under cover	low
C	0.1	eggs laid in relatively deep water	moderate
D	0.2	eggs laid in relatively deep water	moderate
E	0.3	eggs laid in shallow water	high
F	0.5	eggs laid in shallow water	high
G	1.0	eggs laid in shallow water	high

Figure 1 shows the percent of eggs that survived to hatching in the lab for these 7 species after exposure to unfiltered sunlight or to sunlight from which the UV radiation had been filtered out.

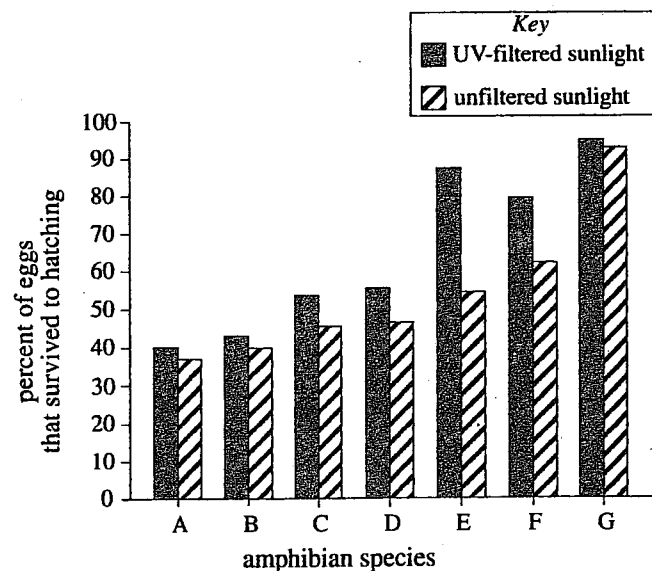


Figure 1

Figure 2 shows predicted UV levels over time in 4 geographic regions that have amphibian populations.

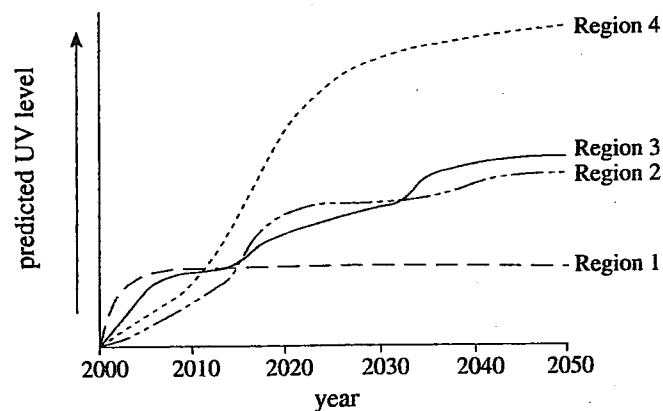


Figure 2



19. Based on the information in Figure 1, eggs from which species are most likely to survive prolonged exposure to sunlight?
- A. Species A
 - B. Species C
 - C. Species F
 - D. Species G
20. According to the data in Figure 1, which species showed the greatest difference between the percent of eggs that survived to hatching after exposure to unfiltered sunlight and the percent of eggs that survived to hatching after exposure to UV-filtered sunlight?
- F. Species A
 - G. Species C
 - H. Species E
 - J. Species G
21. Researchers recently discovered a new amphibian species that lays its eggs under cover. Based on the data in Table 1, the researchers would predict that this species' relative DNA-repair ability is most likely:
- A. less than 0.1.
 - B. greater than 0.1 and less than 0.3.
 - C. greater than 0.3 and less than 0.7.
 - D. greater than 0.7.
22. According to the information in Table 1, for all the species shown, as the exposure of eggs to sunlight increases, the relative ability to repair DNA damage generally:
- F. decreases only.
 - G. increases only.
 - H. decreases, then increases.
 - J. increases, then decreases.
23. Based on the data in Table 1 and Figure 1, amphibians that had the lowest percent of eggs that survived to hatching when exposed to unfiltered sunlight tend to:
- A. bury their eggs.
 - B. lay their eggs under cover.
 - C. lay their eggs in deep water.
 - D. lay their eggs in shallow water.

Passage V

Seashore cliffs often lose rock or sediment to wave erosion. Figure 1 shows cliff compositions; cliff heights, in meters (m); and the net change in mean high water mark (MHW), in m, from 1880 to 1970 along a section of shoreline. A net negative change in MHW indicates a net loss of rock or sediment and a net positive change indicates a net gain of sediment.

Table 1 shows the percentage of a year that vertical sections of a cliff are exposed to wave erosion.

Table 1	
Cliff section height* (m)	Percentage of a year cliff section is exposed to wave erosion
0.0–0.5	52.0
0.5–1.0	37.0
1.0–1.5	21.0
1.5–2.0	9.5
2.0–2.5	3.9
2.5–3.0	1.7
3.0–3.5	0.8
3.5–4.0	0.5

*Note: Heights are measured from mean sea level.

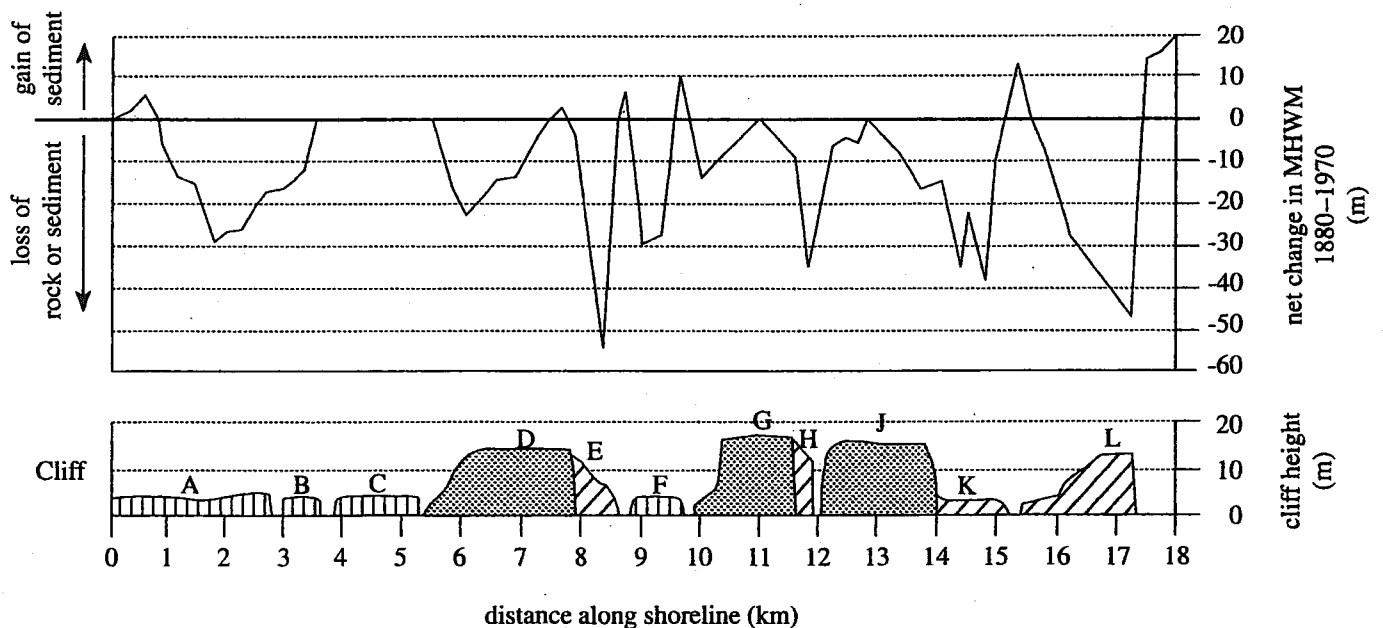
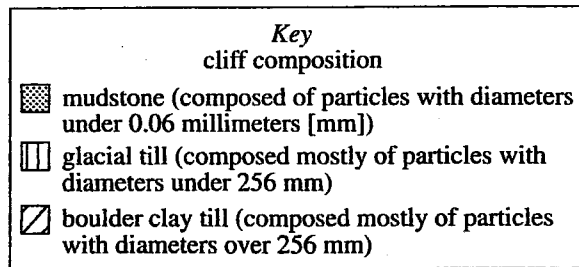


Figure 1

Figure 2 shows Cliff E and F erosion rates, in cm/yr, as they relate to percentage of a year that a cliff section is exposed to wave erosion.

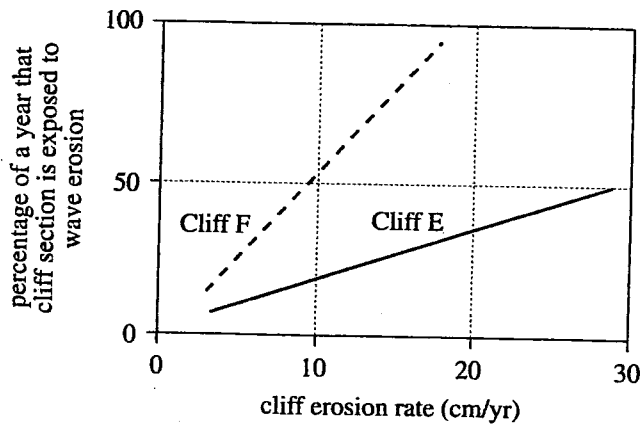


Figure 2

Figures and Table adapted from D. Jones and A. Williams, "Statistical Analysis of Factors Influencing Cliff Erosion Along a Section of the West Wales Coast, U.K." ©1991 by John Wiley and Sons, Ltd.

24. According to Figure 1, at a distance of 12 km along the shoreline, cliffs of what composition are present, if any?

F. Cliffs of glacial till
G. Cliffs of boulder clay till
H. Cliffs of mudstone
J. No cliffs are present.

25. According to the information in Figure 1, one property that was used to distinguish the various materials that compose the cliffs in the study area is the materials':

A. particle diameters.
B. particle density.
C. color.
D. age.

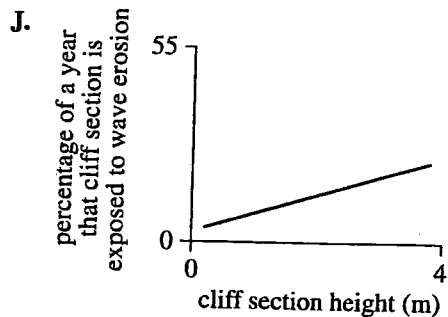
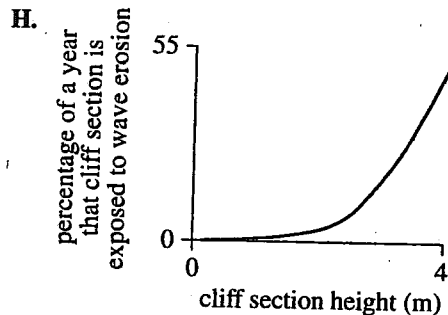
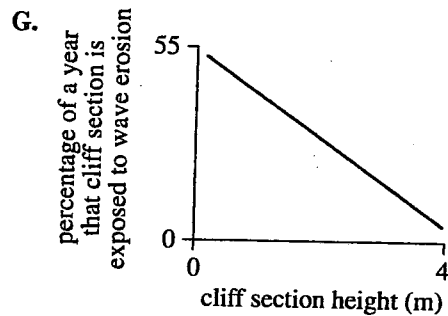
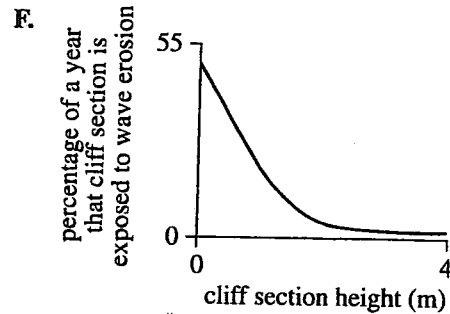
26. Based on the information in Table 1, a cliff section with a height of 4.0–4.5 m above the zero baseline would be exposed to wave erosion approximately what percentage of a year?

F. 10%
G. 2%
H. 0.8%
J. 0.3%

27. According to Figures 1 and 2, the difference between Cliff E and Cliff F erosion rates could best be explained by a difference in the:

A. composition of the 2 cliffs.
B. force of waves on the 2 cliffs.
C. distance of the 2 cliffs along the shoreline.
D. annual rainfall on the 2 cliffs.

28. According to Table 1, which of the following figures best represents the relationship between the height of a cliff section and the percentage of a year that a cliff section is exposed to wave erosion?



Passage VI

Enzymes serve as *catalysts* (compounds that increase the rate of a chemical reaction but are themselves not used up) during digestion. The enzyme *trypsin* is involved in the breakdown of proteins into smaller subunits during digestion. A student investigated the effects of temperature, incubation time, and pH on enzyme activity.

Experiment 1

To each of 11 test tubes, 6 milliliters (mL) of a *casein* (a protein) solution was added. One mL of a trypsin solution was added to each of Tubes 1–10. Tube 11 received 1 mL of water without trypsin. The tubes were then stirred in water baths at various temperatures, and incubated (heated) from 0 to 15 minutes (min). After incubation, 0.1 mL of CaCl_2 solution was added to each tube. CaCl_2 stops the reaction and forms a *precipitate* (solid) with the protein that is not broken down by the trypsin. The precipitates were removed from the tubes and dried. The masses of the precipitates, in milligrams (mg), were measured to determine the relative amount of protein that remained in each tube. The results are in Table 1.

Table 1				
Tube	Temperature of water bath (°C)	Amount of trypsin (mL)	Incubation time (min)	Mass of precipitate (mg)
1	25	1	0	3.0
2	25	1	5	2.4
3	25	1	10	2.0
4	25	1	15	1.7
5	30	1	5	2.4
6	30	1	10	1.4
7	30	1	15	0.5
8	35	1	5	0.1
9	35	1	10	0.1
10	35	1	15	< 0.1
11	35	0	5	3.0

Experiment 2

To each of 6 test tubes, 6 mL of casein solution and 1 mL of trypsin solution were added. Tubes were adjusted to various pHs. Each tube was stirred and incubated at 25°C for 10 min. The relative amount of protein present in each tube was determined as in Experiment 1. The results are in Table 2.

Table 2		
Tube	pH	Mass of precipitate (mg)
12	4	2.8
13	6	2.5
14	8	2.0
15	10	2.3
16	12	2.6
17	14	2.9

29. In Experiment 1, which of the following conditions allowed the large amount of precipitate to form in Tube 11?

- A. Higher temperature
- B. Higher pH
- C. Lack of casein
- D. Lack of trypsin

30. In which of the following ways are the designs of Experiments 1 and 2 different?

- F. A larger volume of trypsin solution per test tube was used in Experiment 1 than in Experiment 2.
- G. Temperature was varied in Experiment 1 but held constant in Experiment 2.
- H. Incubation time remained constant in Experiment 1 but was varied in Experiment 2.
- J. The pH of the solutions in the tubes varied in Experiment 1 but not in Experiment 2.



31. Which of the following hypotheses about the effects of pH on trypsin activity is best supported by the results of Experiment 2 ? As the pH of the solutions increases from 4 to 14, the effectiveness of trypsin:
- A. increases only.
 - B. decreases only.
 - C. increases, then decreases.
 - D. stays the same.
32. Suppose that CaCl_2 had been added immediately to Tube 16 with no incubation time allowed. Based on the results of Experiment 1, one would predict that the amount of precipitate formed would have been approximately:
- F. 2.0 mg.
 - G. 2.3 mg.
 - H. 2.6 mg.
 - J. 3.0 mg.
33. In which of the following tubes was the greatest amount of protein *broken down* by trypsin?
- A. Tube 1
 - B. Tube 3
 - C. Tube 8
 - D. Tube 10
34. According to the results of both experiments, one would predict that the LEAST amount of precipitate would be formed if tubes were incubated for 12 min under which of the following conditions?
- F. 25°C at pH of 4
 - G. 25°C at pH of 8
 - H. 30°C at pH of 4
 - J. 30°C at pH of 8

Passage VII

Some students performed 3 studies to measure the average horizontal speed of a baseball along its flight path under different wind conditions.

In each study, a pitching machine, a catcher, and a camera were positioned as shown in Figure 1. A measuring tape was extended just above the imaginary line from the pitching machine to the catcher.

Study 1

When no wind was blowing, the students caused the pitching machine to pitch the baseball at a speed of 36 meters per second (m/sec). While the baseball was in flight, its image was recorded on the film in the camera every 0.1 sec. The resulting photograph showed 5 images of the baseball below an image of the measuring tape (see Figure 2).

For each interval defined by adjacent images of the baseball, the students measured the horizontal distance between the images (using the image of the measuring tape) and then divided the distance by 0.1 sec to determine the average horizontal speed of the baseball (see Table 1).

Table 1			
Interval	Images	Horizontal distance between images (m)	Average horizontal speed of baseball (m/sec)
A	1 and 2	3.49	34.9
B	2 and 3	3.41	34.1
C	3 and 4	3.34	33.4
D	4 and 5	3.28	32.8

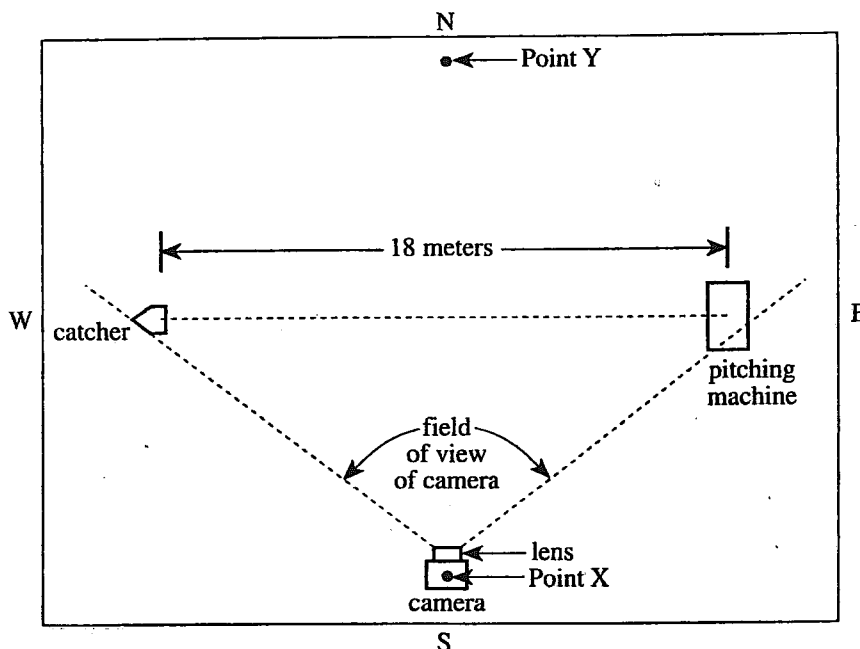


Figure 1

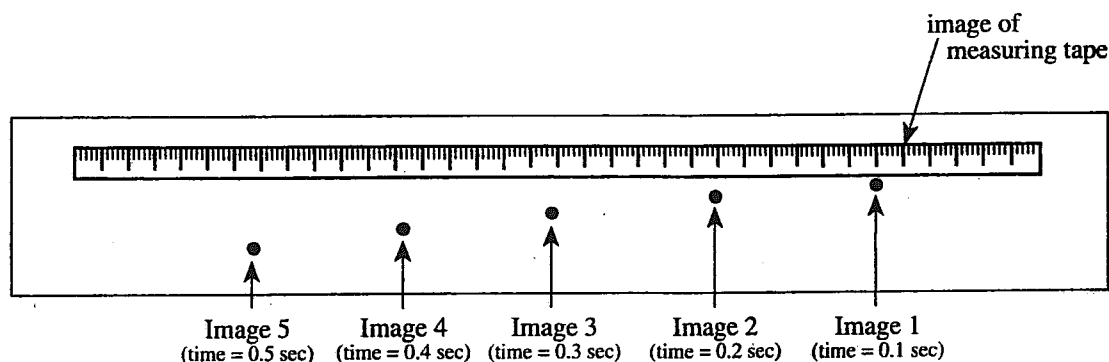


Figure 2

Study 2

When the wind was blowing from east to west with a speed of 15 m/sec, the students repeated the procedure used in Study 1 (see Table 2).

Table 2			
Interval	Images	Horizontal distance between images (m)	Average horizontal speed of baseball (m/sec)
A	1 and 2	3.56	35.6
B	2 and 3	3.53	35.3
C	3 and 4	3.51	35.1
D	4 and 5	3.49	34.9

Study 3

When the wind was blowing from west to east with a speed of 15 m/sec, the students repeated the procedure used in Study 1 (see Table 3).

Table 3			
Interval	Images	Horizontal distance between images (m)	Average horizontal speed of baseball (m/sec)
A	1 and 2	3.37	33.7
B	2 and 3	3.23	32.3
C	3 and 4	3.10	31.0
D	4 and 5	2.97	29.7

35. Based on the results of the studies, the horizontal speed of the baseball upon striking the catcher's mitt was probably greatest in which, if any, study?

- A. Study 1
- B. Study 2
- C. Study 3
- D. None of the studies, because the horizontal speed was the same in all 3 studies.

36. The average horizontal speed of the baseball for any one of the intervals (A, B, C, or D) was different in Studies 2 and 3. This difference was due to:

- F. a difference in wind direction only.
- G. a difference in wind speed only.
- H. a difference in wind direction and a difference in wind speed.
- J. neither a difference in wind direction nor a difference in wind speed.

37. Given that an image of the baseball was recorded on the film in the camera every 0.1 sec, each photograph produced in the studies would have shown fewer than 5 images of the baseball if which of the following had been true?

- A. If the baseball had been pitched at a horizontal speed of 36 m/sec by a pitcher instead of by a pitching machine
- B. If the pitching machine and the catcher had been positioned so that they were 10 m apart rather than 18 m apart
- C. If the measuring tape had been extended just below, rather than just above, the imaginary line from the pitching machine to the catcher
- D. If the camera had been positioned at Point Y in Figure 1 with its lens facing south rather than at Point X in Figure 1 with its lens facing north

38. Suppose that prior to performing a fourth study, the students have reversed the positions of the pitching machine and the catcher. Assuming the baseball is pitched at a speed of 36 m/sec and the wind speed is 15 m/sec, the results of Study 4 will be the same as the results of Study 3 if Study 4 is performed when the wind is blowing from:

- F. east to west.
- G. west to east.
- H. north to south.
- J. south to north.

39. In each study, compared with Interval A, for Interval D, the baseball traveled:

- A. the same horizontal distance in a greater period of time.
- B. the same horizontal distance in a lesser period of time.
- C. a greater horizontal distance in the same period of time.
- D. a lesser horizontal distance in the same period of time.

40. In Studies 2 and 3, it took the baseball 0.51 sec and 0.56 sec, respectively, to travel from the pitching machine to the catcher. If these two studies had been performed when the wind was blowing with a speed of 20 m/sec instead of 15 m/sec, the respective times would probably have been:

- F. greater than 0.51 sec and greater than 0.56 sec.
- G. greater than 0.51 sec and less than 0.56 sec.
- H. less than 0.51 sec and greater than 0.56 sec.
- J. less than 0.51 sec and less than 0.56 sec.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Scoring Your Practice Test

How to Score the Practice Test

The remainder of this booklet provides scoring keys and score conversion tables. Follow the instructions below and on the following pages to score the practice test and review your performance.

Raw Scores

The number of questions you answered correctly on each test and in each subscore area is your raw score. Because there are many forms of the ACT, each containing different questions, some forms will be slightly easier (and some slightly harder) than others. A raw score of 67 on one form of the English Test, for example, may be about as difficult to earn as a raw score of 70 on another form of that test.

To compute your raw scores, check your answers with the scoring keys on pages 57–59. Count the number of correct answers for each of the four tests and seven subscore areas, and enter the number in the blanks provided on those pages. These numbers are your raw scores on the tests and subscore areas.

Scale Scores

To adjust for the small differences that occur among different forms of the ACT, the raw scores for tests and subscore areas are converted into *scale scores*. Scale scores are printed on the reports sent to you and your college and scholarship choices.

When your raw scores are converted into scale scores, it becomes possible to compare your scores with those of examinees who completed different test forms. For example, a scale score of 26 on the English Test has the same meaning regardless of the form of the ACT on which it is based.

To determine the scale scores corresponding to your raw scores on the practice test, use the score conversion tables on pages 60–61. Table 1 on page 60 shows the raw-to-scale score conversions for the total tests, and Table 2 on page 61 shows the raw-to-scale score conversions for the subscore areas. Because each form of the ACT Assessment is unique, each form has somewhat different conversion tables. Consequently, these tables provide only approximations of the raw-to-scale score conversions that would apply if a different form of the ACT Assessment were taken. Therefore, the scale scores obtained from the practice test would not be expected to match precisely the scale scores received from a national administration of the ACT Assessment.

Percent At or Below

Even scale scores don't tell the whole story of your test performance. You may want to know how your scores compare to the scores of other students who take the ACT.

The norms table (Table 3 on page 62) enables you to compare your scores on the sample test with the scores of recent high school graduates who tested as sophomores, juniors, or seniors. The numbers reported in Table 3 are

cumulative percents. A cumulative percent is the percent of students who scored *at or below* a given score. For example, a Composite score of 20 has a cumulative percent of 48. This means that 48% of the ACT-tested high school students had a Composite score of 20 or lower.

Remember that your scores and percent at or below on the sample test are only *estimates* of the scores that you will obtain on an actual form of the ACT. Test scores are only one indicator of your level of academic knowledge and skills. Consider your scores in connection with your grades, your performance in outside activities, and your career interests.

Standards for Transition™

To add to the information you receive about your performance on the ACT Assessment, ACT has developed Standards for Transition. The Standards for Transition help you to more fully understand what your total test score means for each academic area assessed in the ACT Assessment: English, Mathematics, Reading, and Science Reasoning. The Standards for Transition describe the types of skills, strategies, and understandings you will need to make a successful transition from high school to college. Standards are provided for six score ranges that reflect the progression and complexity of skills in the four academic areas measured in the ACT Assessment. The Standards for Transition can be found at ACT's website (www.act.org) and in the student guide, *Using Your ACT Assessment Results*, that you will receive with your score report.

Reviewing Your Performance on the Practice Test

After you have determined your scale scores, consider the following as you evaluate how you did on the practice test.

- Did you run out of time before you completed a test? If so, reread the information in this booklet on pacing yourself. Perhaps you need to adjust the way you used your time in responding to the questions. It is to your advantage to answer every question and pace yourself so that you can do so. Remember there is no penalty for guessing.
- Did you spend too much time trying to understand the directions to the tests? If so, read the directions for each test again thoroughly. The directions in the practice test are exactly like the directions that will appear in your test booklet on the test day. Make sure you understand them now, so you won't have to spend too much time studying them when you take the actual test.
- Review the questions that you missed. Did you select a response that was an incomplete answer or that did not directly respond to the question being asked? Try to figure out what you overlooked in answering the questions.
- Did a particular type of question confuse you? Did the questions you missed come from a particular subscore area? In reviewing your responses to the practice test, check to see whether a particular type of question or a particular subscore area was more difficult for you or took more of your time.

Scoring Keys for the ACT Practice Test

Use the scoring key for each test to score your answer document for the practice test. Mark a "1" in the blank for each question you answered correctly. Add up the numbers in each subscore area and enter the total number correct for each subscore area in the blanks provided. Also enter the total number correct for each test in the blanks provided. The total number correct for each test is the sum of the number correct in each subscore area.

Test 1: English—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	UM	RH	Key	UM	RH	Key	UM	RH
1. D			26. H			51. A		
2. G			27. B			52. H		
3. D			28. F			53. D		
4. F			29. B			54. J		
5. B			30. F			55. A		
6. F			31. A			56. J		
7. B			32. F			57. C		
8. G			33. C			58. G		
9. C			34. G			59. D		
10. J			35. D			60. H		
11. C			36. G			61. A		
12. J			37. B			62. F		
13. C			38. G			63. D		
14. F			39. D			64. G		
15. A			40. H			65. C		
16. J			41. D			66. F		
17. B			42. F			67. A		
18. G			43. C			68. H		
19. B			44. J			69. B		
20. J			45. A			70. G		
21. C			46. G			71. A		
22. J			47. B			72. G		
23. C			48. H			73. D		
24. J			49. A			74. G		
25. C			50. H			75. A		

Number Correct (Raw Score) for:	
Usage/Mechanics (UM) Subscore Area	(40)
Rhetorical Skills (RH) Subscore Area	(35)
Total Number Correct for English Test (UM + RH)	(75)

* UM = Usage/Mechanics
RH = Rhetorical Skills

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Test 2: Mathematics—Scoring Key

Subscore Area*				Subscore Area*			
Key	EA	AG	GT	Key	EA	AG	GT
1. E				31. D			
2. J				32. G			
3. A				33. E			
4. G				34. J			
5. D				35. E			
6. H				36. K			
7. D				37. E			
8. G				38. G			
9. B				39. E			
10. H				40. H			
11. D				41. D			
12. K				42. G			
13. C				43. B			
14. F				44. F			
15. E				45. B			
16. G				46. H			
17. C				47. A			
18. G				48. G			
19. A				49. C			
20. K				50. K			
21. A				51. D			
22. J				52. G			
23. D				53. C			
24. H				54. F			
25. D				55. C			
26. H				56. F			
27. B				57. D			
28. H				58. K			
29. D				59. A			
30. F				60. F			

Number Correct (Raw Score) for:

Pre-Alg./Elem. Alg. (EA) Subscore Area	(24)
Inter. Alg./Coord. Geo. (AG) Subscore Area	(18)
Plane Geo./Trig. (GT) Subscore Area	(18)
Total Number Correct for Math Test (EA + AG + GT)	(60)

* EA = Pre-Algebra/Elementary Algebra
AG = Intermediate Algebra/Coordinate Geometry
GT = Plane Geometry/Trigonometry

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Test 3: Reading—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	SS	AL	Key	SS	AL	Key	SS	AL
1. A		_____	15. B		_____	29. D		_____
2. G		_____	16. G		_____	30. H		_____
3. C		_____	17. D		_____	31. C		_____
4. H		_____	18. H		_____	32. J		_____
5. D		_____	19. C		_____	33. A		_____
6. G		_____	20. J		_____	34. G		_____
7. D		_____	21. D		_____	35. B		_____
8. F		_____	22. H		_____	36. J		_____
9. A		_____	23. C		_____	37. C		_____
10. G		_____	24. G		_____	38. J		_____
11. B	_____		25. A		_____	39. B		_____
12. H	_____		26. H		_____	40. F		_____
13. D	_____		27. A		_____			
14. F	_____		28. G		_____			

Number Correct (Raw Score) for:

Social Studies/Sciences (SS) Subscore Area _____
(20)

Arts/Literature (AL) Subscore Area _____
(20)

Total Number Correct for Reading Test (SS + AL) _____
(40)

* SS = Social Studies/Sciences
AL = Arts/Literature

Test 4: Science Reasoning—Scoring Key

Key		Key		Key	
1. A	_____	15. B	_____	29. D	_____
2. F	_____	16. H	_____	30. G	_____
3. C	_____	17. A	_____	31. C	_____
4. H	_____	18. G	_____	32. J	_____
5. D	_____	19. D	_____	33. D	_____
6. J	_____	20. H	_____	34. J	_____
7. B	_____	21. A	_____	35. B	_____
8. G	_____	22. G	_____	36. F	_____
9. C	_____	23. A	_____	37. B	_____
10. F	_____	24. J	_____	38. F	_____
11. C	_____	25. A	_____	39. D	_____
12. G	_____	26. J	_____	40. H	_____
13. A	_____	27. A	_____		
14. J	_____	28. F	_____		

Number Correct (Raw Score) for:

Total Number Correct for Science Reasoning Test _____
(40)

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TABLE 1

Procedures Used to Obtain Scale Scores From Raw Scores for the ACT Practice Test

On each of the four tests on which you marked any responses, the total number of correct responses yields a raw score. Use the table below to convert your raw scores to scale scores. For each test, locate and circle your raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale score that corresponds to that raw score. As you determine your scale scores, enter them in the blanks provided on the right. The highest possible scale score for each test is 36. The lowest possible scale score for any test on which you marked any response is 1.

Next, compute the Composite score by averaging the four scale scores. To do this, add your four scale scores and divide the sum by 4. If the resulting number ends in a fraction, round it off to the nearest whole number. (Round down any fraction less than one-half; round up any fraction that is one-half or more.) Enter this number in the blank. This is your Composite score. The highest possible Composite score is 36. The lowest possible Composite score is 1.

ACT Test	Your Scale Score
English	_____
Mathematics	_____
Reading	_____
Science Reasoning	_____
Sum of scores	_____
Composite score (sum ÷ 4)	_____

NOTE: If you left a test completely blank and marked no items, do not list a scale score for that test. If any test was completely blank, do not calculate a Composite score.

Scale Score	Raw Scores				Scale Score
	Test 1 English	Test 2 Mathematics	Test 3 Reading	Test 4 Science Reasoning	
36	75	60	39-40	40	36
35	74	—	38	—	35
34	73	59	37	39	34
33	71-72	58	—	—	33
32	70	57	36	38	32
31	69	55-56	35	37	31
30	68	53-54	34	—	30
29	66-67	52	—	36	29
28	64-65	50-51	33	35	28
27	62-63	48-49	32	34	27
26	60-61	45-47	31	33	26
25	58-59	43-44	30	31-32	25
24	55-57	41-42	29	30	24
23	53-54	39-40	27-28	28-29	23
22	50-52	37-38	26	26-27	22
21	48-49	35-36	25	25	21
20	45-47	32-34	23-24	23-24	20
19	42-44	29-31	22	21-22	19
18	39-41	26-28	21	19-20	18
17	37-38	22-25	20	17-18	17
16	34-36	19-21	18-19	15-16	16
15	31-33	15-18	17	14	15
14	29-30	12-14	15-16	12-13	14
13	27-28	10-11	13-14	10-11	13
12	25-26	08-09	10-12	09	12
11	23-24	06-07	08-09	08	11
10	20-22	05	07	06-07	10
9	18-19	04	06	05	9
8	15-17	—	05	04	8
7	12-14	03	—	03	7
6	10-11	—	04	—	6
5	07-09	02	03	02	5
4	05-06	—	02	—	4
3	04	01	—	01	3
2	02-03	—	01	—	2
1	00-01	00	00	00	1

TABLE 2

Procedures Used to Obtain Scale Subscores From Raw Scores for the ACT Practice Test

For each of the seven subscore areas, the total number of correct responses yields a raw score. Use the table below to convert your raw score to scale subscores. For each of the seven subscore areas, locate and circle either the raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale subscore that corresponds to that raw score. As you determine your scale subscores, enter them in the blanks provided on the right. The highest possible scale subscore is 18. The lowest possible scale subscore is 1.

If you left a test completely blank and marked no items, do not list any scale subscores for that test.

ACT Test	Your Scale Subscore
English	
Usage/Mechanics (UM)	
Rhetorical Skills (RH)	
Mathematics	
Pre-Algebra/Elem. Algebra (EA)	
Inter. Algebra/Coord. Geometry (AG)	
Plane Geometry/Trigonometry (GT)	
Reading	
Social Studies/Sciences (SS)	
Arts/Literature (AL)	

Scale Subscore	Raw Scores							Scale Subscore
	Test 1 English		Test 2 Mathematics			Test 3 Reading		
	Usage/ Mechanics	Rhetorical Skills	Pre-Algebra/ Elem. Algebra	Inter. Algebra/ Coord. Geometry	Plane Geometry/ Trigonometry	Social Studies/ Sciences	Arts/ Literature	
18	39-40	35	24	18	18	20	20	18
17	38	33-34	23	17	—	18-19	19	17
16	36-37	32	—	—	16-17	17	18	16
15	35	30-31	22	15-16	15	16	17	15
14	33-34	29	21	14	13-14	15	16	14
13	31-32	27-28	20	13	12	14	—	13
12	29-30	25-26	18-19	11-12	10-11	13	15	12
11	27-28	22-24	16-17	10	08-09	12	14	11
10	24-26	20-21	15	08-09	07	10-11	13	10
9	22-23	17-19	12-14	07	05-06	09	12	9
8	19-21	14-16	10-11	05-06	04	08	11	8
7	17-18	12-13	08-09	—	03	06-07	10	7
6	15-16	11	06-07	04	02	05	08-09	6
5	12-14	09-10	04-05	03	—	04	06-07	5
4	10-11	06-08	03	02	01	03	05	4
3	07-09	04-05	02	—	—	02	03-04	3
2	05-06	02-03	01	01	—	01	02	2
1	00-04	00-01	00	00	00	00	00-01	1

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✂ ACT English Practice Tests:

<https://www.crackab.com/act/english/>

✂ ACT Math Practice Tests:

<https://www.crackab.com/act/math/>

✂ ACT Reading Practice Tests:

<https://www.crackab.com/act/reading/>

✂ ACT Science Practice Tests:

<https://www.crackab.com/act/science/>

ACT Grammar: <https://www.crackab.com/act/grammar/>

ACT Real Past Papers Download:

<https://www.crackab.com/act-downloads/>

Digital SAT & New SAT Practice Tests:

<https://www.cracksat.net>

Real SAT Tests Download:

<http://www.cracksat.net/sat-downloads/>

AP Exams Practice Tests:

<https://www.crackap.com>

<https://www.apstudy.net>

TABLE 3
Norms Table for the ACT Practice Test

Use the norms table below to determine your estimated percent at or below for each of your scale scores.

In the far left column, circle your scale score for the English Test (from page 60). Then read across to the percent at or below column for that test; circle or put a check mark beside the corresponding percent at or below. Use the same procedure for each test (from page 60) and subscore area (from page 61). You may find it easier to use the right column of scale scores for your Science Reasoning Test and Composite scores.

As you mark your percents at or below, enter them in the blanks provided at the right.

You may also find it helpful to compare your performance with the national mean (average) score for each of the four tests, subscore areas, and the Composite as shown at the bottom of the norms table.

ACT Test	Your Estimated Percent At or Below on Practice Test
English	
Usage/Mechanics	
Rhetorical Skills	
Mathematics	
Pre-Algebra/Elem. Alg.	
Alg./Coord. Geometry	
Plane Geometry/Trig.	
Reading	
Soc. Studies/Sciences	
Arts/Literature	
Science Reasoning	
Composite	

National Distributions of "Percent At or Below" for ACT Test Scores 2001 HS Graduates Tested on National Test Dates													
Score	ENGLISH			MATHEMATICS				READING	Soc. Studies/Sciences		SCIENCE REASONING	COMPOSITE	Score
	Usage/Mechanics	Rhetorical Skills		Pre-Algebra/Elem. Alg.	Alg./Coord. Geometry	Plane Geometry/Trig.		Arts/Literature					
36	99			99			99			99	99	99	36
35	99			99			99			99	99	99	35
34	99			99			98			99	99	99	34
33	99			99			97			99	99	99	33
32	98			99			96			99	99	99	32
31	97			98			94			98	98	98	31
30	96			96			92			97	97	97	30
29	94			94			89			96	95	95	29
28	92			92			87			94	93	93	28
27	88			89			83			91	90	90	27
26	85			86			79			88	86	86	26
25	81			80			75			85	82	82	25
24	76			76			70			78	76	76	24
23	70			71			65			72	70	70	23
22	65			66			59			65	64	64	22
21	58			61			52			56	56	56	21
20	52			55			47			48	48	48	20
19	45			48			40			38	40	40	19
18	37	99	99	41	99	99	35	99	99	30	33	33	18
17	31	99	99	32	97	99	29	99	97	21	25	25	17
16	25	95	98	22	93	98	24	94	91	15	18	18	16
15	20	91	95	14	89	96	20	90	85	10	13	13	15
14	15	86	89	07	84	93	15	84	77	07	08	08	14
13	11	80	82	03	76	87	11	75	71	04	04	04	13
12	08	72	75	01	68	80	06	68	64	02	02	02	12
11	06	66	64	01	59	67	02	60	55	01	01	01	11
10	04	56	50	01	47	59	01	51	46	01	01	01	10
09	02	44	38	01	39	41	01	37	37	01	01	01	09
08	01	35	24	01	27	26	01	27	29	01	01	01	08
07	01	23	14	01	17	14	01	18	22	01	01	01	07
06	01	15	08	01	09	10	01	11	16	01	01	01	06
05	01	10	04	01	04	06	01	06	09	01	01	01	05
04	01	05	02	01	01	03	01	03	04	01	01	01	04
03	01	02	01	01	01	01	01	01	01	01	01	01	03
02	01	01	01	01	01	01	01	01	01	01	01	01	02
01	01	01	01	01	01	01	01	01	01	01	01	01	01
Mean	20.5	10.2	10.6	20.7	10.9	10.2	21.3	10.8	11.0	21.0	21.0		
S.D.	5.6	3.5	2.9	5.0	3.4	2.9	6.0	3.5	3.9	4.6	4.7		
Note: These norms are the source of national and state norms printed on ACT Assessment score reports during the 2001-2002 testing year. Sample size: 1,069,772.													